

# **ESAT CAMP MOCK C**

Full sitting

EC-FULL-C

**MATHEMATICS 1**

**40 MINUTES**

**MATHEMATICS 2**

**40 MINUTES**

**PHYSICS**

**40 MINUTES**

**CHEMISTRY**

**40 MINUTES**

**BIOLOGY**

**40 MINUTES**

## **INSTRUCTIONS TO CANDIDATES**

Please read these instructions carefully, but **do not open this question paper until you are ready to begin**.

This paper contains **5 modules** (Mathematics 1, Mathematics 2, Physics, Chemistry, Biology). Each module has **27 multiple-choice questions** and is designed for **40 minutes** (about **200 minutes** in total if you sit every module).

There are no penalties for incorrect responses, only marks for correct answers, so you should attempt all of the questions. Each question is worth one mark.

For each question, choose the one option you consider correct. If you make a mistake, erase thoroughly and try again.

You can use the question paper for rough working. Dictionaries and calculators are **NOT permitted**.

**Please wait until you are ready before turning this page.**

*Original ESAT CAMP practice material. Not an official UAT-UK or Pearson paper.*

ESAT CAMP · Independent preparation resource

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## **Mathematics 1**

- 1 A dataset of  $2n$  numbers is partitioned into two groups of  $n$  numbers. Two numbers are then removed from the dataset.

The table below summarises the number of values and the mean for each group, as well as for the removed and remaining values:

| Values           | Number of values | Mean |
|------------------|------------------|------|
| First group      | $n$              | $a$  |
| Second group     | $n$              | $b$  |
| Removed values   | 2                | $c$  |
| Remaining values | $2n-2$           | $m$  |

Which of the following is a correct expression for  $n$  in terms of  $a$ ,  $b$ ,  $c$ , and  $m$ ?

- A  $\frac{2(c-m)}{a+b-2m}$
- B  $\frac{2(c-m)}{a+b-m}$
- C  $\frac{c-m}{a+b-2m}$
- D  $\frac{2(m-c)}{a+b-2m}$
- E  $\frac{2(c+m)}{a+b-2m}$
- F  $\frac{2(c-m)}{a+b+2m}$

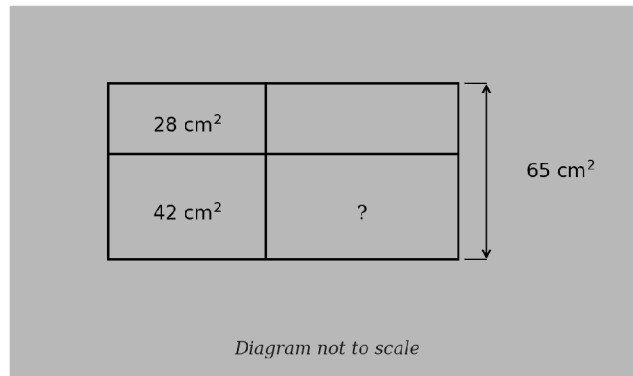
- 2 A rectangle is divided into four smaller non-overlapping rectangles by two straight lines parallel to its sides, as shown in the diagram.

The area of the top-left rectangle is  $28 \text{ cm}^2$ .

The area of the bottom-left rectangle is  $42 \text{ cm}^2$ .

The combined area of the two rectangles on the right is  $65 \text{ cm}^2$ .

What is the area of the bottom-right rectangle?



- A  $26 \text{ cm}^2$   
 B  $33 \text{ cm}^2$   
 C  $39 \text{ cm}^2$   
 D  $45 \text{ cm}^2$   
 E  $51 \text{ cm}^2$   
 F  $81 \text{ cm}^2$
- 3 The curve with equation

$$y = \log_2(x)$$

meets the curve with equation

$$y = 3 - \log_2(k - x)$$

where  $k$  is a positive constant, at two distinct points.

The  $x$ -coordinates of these two points are  $p$  and  $q$ , where  $q > p$ .

Given that  $q - p = 2$ , what is the value of  $k$ ?

- A 4  
 B  $2\sqrt{5}$   
 C  $4\sqrt{2}$   
 D 6  
 E 8  
 F  $2\sqrt{10}$

- 4** The positive integers are written in a grid with 6 columns, numbered 1 to 6 from left to right.  
 The first row contains the integers 1 to 6, placed in columns 1 to 6 respectively.  
 The second row contains the integers 7 to 12, placed in columns 6 down to 1 respectively.  
 The third row contains the integers 13 to 18, placed in columns 1 to 6 respectively.  
 This alternating pattern continues indefinitely.

In which column is the integer 250?

- A** 1
- B** 2
- C** 3
- D** 4
- E** 5
- F** 6

- 5** A box contains 5 fair six-sided dice and 3 biased six-sided dice.

For each biased die, the probability of rolling a 6 is  $\frac{1}{2}$ , and the numbers 1 to 5 are equally likely.

A die is chosen at random from the box and is rolled twice.

Given that the first roll shows a 6, what is the probability that the second roll also shows a 6?

- A**  $\frac{1}{9}$
- B**  $\frac{1}{6}$
- C**  $\frac{7}{24}$
- D**  $\frac{1}{3}$
- E**  $\frac{8}{21}$
- F**  $\frac{1}{2}$
- G**  $\frac{9}{14}$

- 6** A car travels a distance of  $D$  miles. The fuel consumption of the car is  $F$  litres per 100 kilometres. Given that 5 miles is equivalent to 8 kilometres, which of the following expressions represents the total volume of fuel consumed, in litres?
- A**  $\frac{2DF}{125}$
- B**  $\frac{DF}{160}$
- C**  $\frac{8DF}{5}$
- D**  $\frac{5DF}{8}$
- E**  $160DF$
- F**  $\frac{125DF}{2}$
- 7** A straight line  $L$  passes through the origin and the point  $(4, 12)$ . A point  $P$  in the first quadrant is at a distance of 18 units from the origin. The line segment connecting the origin to  $P$  makes an angle of  $30^\circ$  with the positive  $x$ -axis. What is the sum of the gradient of  $L$  and the  $y$ -coordinate of  $P$ ?
- A** 9
- B** 12
- C** 21
- D**  $\frac{28}{3}$
- E**  $3 + 9\sqrt{3}$
- F**  $\frac{1}{3} + 9\sqrt{3}$
- 8** Two fair, six-sided dice are thrown. Given that the sum of the numbers shown is at least 9, what is the probability that the two numbers are equal?
- A**  $1/6$
- B**  $1/5$
- C**  $5/18$
- D**  $1/3$
- E**  $3/5$

- 9** Water flows from a cylindrical pipe with an internal diameter of 200 mm into an open rectangular tank. The base of the tank is a square with side length 10 cm. The water flows through the pipe at a constant velocity of 1.2 m/s.

What is the rate of increase of the depth of the water in the tank, in cm/min?

- A**  $2\pi$
- B**  $8\pi$
- C**  $720\pi$
- D**  $7200\pi$
- E**  $28800\pi$
- F**  $72000\pi$

- 10** A particle undergoes symmetric, periodic oscillation about a central point. The maximum deviation of the particle from the central point is 10 cm, and the duration of one complete cycle is 0.50 s.

What is the total distance traveled by the particle in a time interval of 3.0 s?

- A** 0 cm
- B** 40 cm
- C** 60 cm
- D** 120 cm
- E** 240 cm

- 11** A solid metal sphere of diameter 6 cm and a solid metal cylinder of diameter 8 cm and height 3 cm are melted down. The molten metal is used to cast a number of identical solid cones, each with diameter 2 cm and height 3 cm.

Assuming no metal is lost, how many complete cones can be made?

- A** 28
- B** 36
- C** 48
- D** 75
- E** 84
- F** 120

- 12** A number of coin flips,  $N$ , is chosen with equal probability from the set  $\{2, 3, 4\}$ . A fair coin is then flipped  $N$  times.

What is the probability that exactly two Heads are observed?

- A**  $\frac{1}{4}$
- B**  $\frac{5}{14}$
- C**  $\frac{7}{48}$
- D**  $\frac{1}{3}$
- E**  $\frac{3}{8}$
- F**  $\frac{5}{8}$
- G**  $\frac{1}{6}$
- H**  $\frac{5}{12}$

- 13** A machine extrudes a solid cylindrical rod of plastic. The plastic flows out at a constant rate of  $0.072\pi \text{ m}^3$  per hour. The diameter of the extruded rod is constant at 40 mm.

What is the linear speed at which the rod is produced, in cm/s?

- A** 0.05
- B** 0.5
- C** 1.25
- D** 5
- E** 75
- F** 300

- 14** A company purchases large quantities of three raw materials. The prices of these materials are  $p$ ,  $q$ , and  $r$  dollars per tonne, respectively.

The company spends an equal amount of money on each of the three materials.

What is the average price, in dollars per tonne, that the company pays for all the materials purchased?

- A**  $\frac{p+q+r}{3}$   
**B**  $\frac{3}{p+q+r}$   
**C**  $\frac{3pqr}{p+q+r}$   
**D**  $\frac{3pqr}{pq+qr+rp}$   
**E**  $\frac{pq+qr+rp}{3pqr}$   
**F**  $\frac{p^2+q^2+r^2}{p+q+r}$

- 15** A point P moves anticlockwise on a circle with equation  $x^2 + y^2 = R^2$ , where  $R > 0$ . It starts at the point  $(R, 0)$  and moves with a constant period of  $T$  for one full revolution.

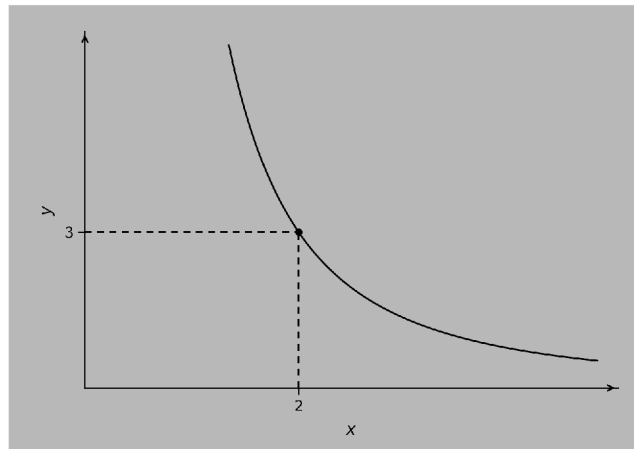
A second point, Q, starts at  $(R, H)$ , where  $H > 0$ , and moves vertically downwards with a constant speed  $v$ .

The two points start moving at the same time. Q reaches the  $x$ -axis at the exact same instant that P reaches the positive  $y$ -axis for the first time.

What is  $v$  in terms of  $H$  and  $T$ ?

- A**  $\frac{4H}{T}$   
**B**  $\frac{2H}{T}$   
**C**  $\frac{H}{T}$   
**D**  $\frac{HT}{4}$   
**E**  $\frac{4R}{T}$   
**F**  $\frac{2H}{\pi T}$

- 16** The diagram shows the graph of  $y$  against  $x$  for  $x > 0$ , where  $y$  is inversely proportional to  $x^2$ .  
The variable  $x$  is directly proportional to  $z^3$ . When  $z = 2$ ,  $x = 4$ .  
Which of the following expressions gives  $y$  in terms of  $z$ ?

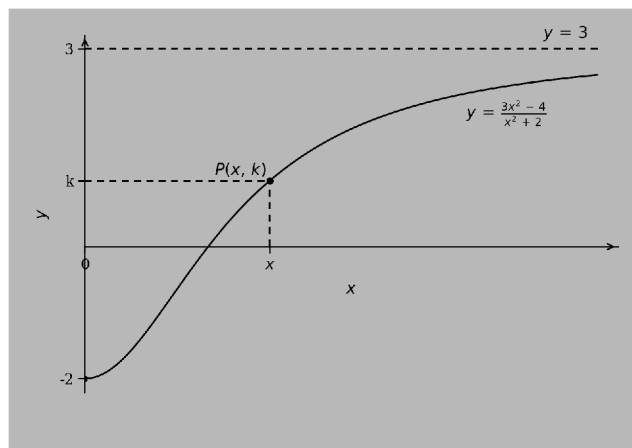


- |          |                      |
|----------|----------------------|
| <b>A</b> | $y = \frac{3}{z^6}$  |
| <b>B</b> | $y = \frac{24}{z^6}$ |
| <b>C</b> | $y = \frac{48}{z^6}$ |
| <b>D</b> | $y = \frac{48}{z^5}$ |
| <b>E</b> | $y = \frac{24}{z^3}$ |
| <b>F</b> | $y = \frac{48}{z^3}$ |
| <b>G</b> | $y = \frac{z^6}{48}$ |
| <b>H</b> | $y = \frac{z^6}{12}$ |
- 17** Two vertices of a regular hexagon are located at  $(-1, 2)$  and  $(3, 6)$ . What is the difference between the areas of the largest and smallest possible regular hexagons that can be drawn with these points as two of their vertices?
- |          |              |
|----------|--------------|
| <b>A</b> | $4\sqrt{3}$  |
| <b>B</b> | $12\sqrt{3}$ |
| <b>C</b> | $24\sqrt{3}$ |
| <b>D</b> | $32\sqrt{3}$ |
| <b>E</b> | $36\sqrt{3}$ |
| <b>F</b> | $48\sqrt{3}$ |
| <b>G</b> | $60\sqrt{3}$ |
| <b>H</b> | $64\sqrt{3}$ |

- 18 A set contains the integers from 1 to  $n$ , where  $n \geq 2$ .  
Two distinct integers are chosen at random from the set.  
The probability that their sum is odd is  $\frac{12}{23}$ .  
What is the sum of all possible values of  $n$ ?

A 23  
B 24  
C 35  
D 46  
E 47  
F 48

- 19 The diagram shows the graph of  $y = \frac{3x^2 - 4}{x^2 + 2}$  for  $x \geq 0$ , with a horizontal asymptote at  $y = 3$  and a  $y$ -intercept at  $(0, -2)$ .  
A horizontal line  $y = k$ , where  $-2 < k < 3$ , intersects the curve at the point  $P(x, k)$ .  
Which of the following is an expression for  $x$  in terms of  $k$ ?



A  $x = \sqrt{\frac{2k+4}{3-k}}$   
B  $x = \sqrt{\frac{2k+4}{3+k}}$   
C  $x = \sqrt{\frac{4-2k}{3-k}}$   
D  $x = \sqrt{\frac{4-2k}{3+k}}$   
E  $x = \sqrt{\frac{k+4}{3-k}}$   
F  $x = \sqrt{\frac{3-k}{2k+4}}$

- 20** The table below shows the frequency distribution of a data set, where  $x$  is a positive integer.

| Value | Frequency |
|-------|-----------|
| 1     | 4         |
| 2     | $x$       |
| 3     | 7         |
| 4     | $2x$      |
| 5     | 5         |

The mean of the data set is 0.2 greater than the median of the data set.

What is the value of  $x$ ?

- A** 2
  - B** 3
  - C** 4
  - D** 5
  - E** 6
- 21** A circle is tangent to the line  $y = 0$  and the line  $3x - 4y = 0$ . The circle passes through the point  $(6, 3)$ .  
What is the sum of the possible values for the radius of the circle?
- A** 3
  - B**  $14/3$
  - C** 5
  - D** 15
  - E** 18

- 22** A dataset consists of  $n$  numbers, where  $n$  is an odd integer greater than 1. The mean, median, and range of the dataset are denoted by  $\mu$ ,  $m$ , and  $r$  respectively.

A single new number with value  $m$  is added to the dataset.

Let  $\mu'$ ,  $m'$ , and  $r'$  denote the mean, median, and range of the new dataset.

Which of the following statements must be true?

1.  $m' = m$  2.  $r' = r$  3.  $\mu' = \mu$

- A** 1 only  
**B** 2 only  
**C** 3 only  
**D** 1 and 2 only  
**E** 1 and 3 only  
**F** 2 and 3 only  
**G** 1, 2 and 3  
**H** None of the statements
- 23** The coefficients  $B$  and  $C$  of the quadratic equation  $x^2 + Bx + C = 0$  are determined as follows.  
 A fair coin is tossed two times, and  $B$  is the number of heads. A second fair coin is tossed one time, and  $C$  is the number of heads.

What is the probability that the equation has at least one real root?

- A**  $\frac{1}{3}$   
**B**  $\frac{3}{8}$   
**C**  $\frac{1}{2}$   
**D**  $\frac{5}{8}$   
**E**  $\frac{2}{3}$   
**F**  $\frac{7}{8}$

- 24** A function  $F$  is constructed by composing four functions in a sequence:

$$F(x) = f_4(f_3(f_2(f_1(x))))$$

Each of the functions  $f_1, f_2, f_3, f_4$  is chosen independently from a set containing 3 distinct functions.

Given that any two different sequences of choices result in a distinct function  $F$ , how many possible functions  $F$  can be constructed?

- A** 12
- B** 15
- C** 24
- D** 27
- E** 64
- F** 81

- 25** A function  $f$  is defined for  $x > 0$  by

$$f(x) = a\sqrt{x} + \frac{b}{\sqrt{x}}$$

where  $a$  and  $b$  are constants.

Given that  $f(1) = 16$  and  $f(4) = 14$ , what is the value of  $f(16)$ ?

- A** 6
- B** 10
- C** 14
- D** 19
- E** 28

- 26** The positive variables  $p, q, r, k, x, y$  are related by the equations:

$$p = qr$$

$$r = \frac{kx}{y}$$

Given that  $p = 6.0 \times 10^1$ ,  $q = 3.0 \times 10^{-2}$ ,  $k = 4.0 \times 10^{-5}$ , and  $y = 4.0 \times 10^{-8}$ , what is the value of  $x$ ?

- A** 0.2
- B** 0.5
- C** 2.0
- D** 20
- E**  $1.8 \times 10^{-3}$
- F**  $2.0 \times 10^6$

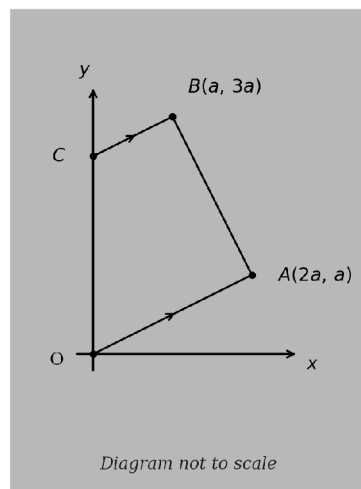
- 27** A circular sector has a total area of  $18\pi$ . The area of the sector per unit of arc length is 3. What is the arc length of the sector?
- A**  $3\pi$
  - B** 6
  - C**  $6\pi$
  - D**  $12\pi$
  - E**  $54\pi$

## **Mathematics 2**

- 1 Find the sum of all solutions  $\theta$  in the interval  $0 \leq \theta < 2\pi$  to the equation:

$$\frac{2 \sin^2 \theta + \sin \theta - 1}{\sin \theta + 1} = 0$$

- A  $\frac{\pi}{6}$   
 B  $\pi$   
 C  $\frac{3\pi}{2}$   
 D  $\frac{5\pi}{2}$   
 E  $3\pi$   
 F 0
- 2 The diagram shows a trapezium  $OABC$  in the first quadrant ( $x \geq 0, y \geq 0$ ).  
 The vertex  $O$  is at the origin  $(0, 0)$ , vertex  $A$  has coordinates  $(2a, a)$ , vertex  $B$  has coordinates  $(a, 3a)$ , and vertex  $C$  lies on the positive  $y$ -axis, where  $a > 0$ .  
 The side  $CB$  is parallel to the side  $OA$ .  
 The area of the trapezium  $OABC$  is 60 square units.  
 What are the coordinates of vertex  $B$ ?



- A  $(2, 6)$   
 B  $(2\sqrt{2}, 6\sqrt{2})$   
 C  $(2\sqrt{5}, 6\sqrt{5})$   
 D  $(4, 12)$   
 E  $(6, 18)$   
 F  $(8, 4)$

- 3 An arithmetic progression has first term 100 and common difference -6.  
What is the maximum possible value of the sum of the first  $n$  terms?

A 4  
B 850  
C 880  
D 882  
E 884  
F 901

- 4 A curve passes through the origin and for  $x > 0$  its gradient is given by

$$\frac{dy}{dx} = 3x^2$$

At a point  $P$  on the curve, the gradient is  $m$ , where  $m$  is a positive constant.  
What is the  $y$ -coordinate of  $P$  in terms of  $m$ ?

A  $\left(\frac{m}{3}\right)^{\frac{3}{2}}$   
B  $\frac{m^3}{27}$   
C  $\left(\frac{3}{m}\right)^{\frac{3}{2}}$   
D  $\frac{m^2}{12}$   
E  $m$   
F  $\left(\frac{m^3}{3}\right)^{\frac{1}{2}}$

- 5 A monic cubic curve  $y = f(x)$  has stationary points at  $x = 1$  and  $x = 3$ .

Which one of the following conditions is **not** sufficient to determine the equation of the curve uniquely?

A The curve passes through the point (2, 5).  
B The  $y$ -coordinate of the local maximum is 1.  
C The curve passes through the point (2, 1).  
D The value of  $\int_0^1 f(x) dx$  is 4.  
E The equation  $f(x) = 0$  has exactly two distinct real roots.  
F The tangent to the curve at  $x = 0$  passes through the point (1, 5).

- 6 A quantity is modelled by a function  $f(x)$  that exhibits exponential decay. For a fixed positive constant  $d$ , increasing  $x$  by  $d$  causes the value of  $f(x)$  to decrease by 20%.

What is the total percentage decrease in the value of  $f(x)$  when  $x$  is increased by  $2d$ ?

- A 36%
- B 40%
- C 64%
- D 4%
- E 44%
- F 16%

- 7 Variables  $x$  and  $y$  are related by the equation  $y = Ab^{2x+3}$ , where  $A$  and  $b$  are constants. A graph of  $\ln y$  against  $x$  is a straight line with gradient 4 and intercept 7 on the vertical axis. Which of the following is the value of  $A$ ?

- A  $e^{-5}$
- B  $e$
- C  $e^2$
- D  $e^4$
- E  $e^7$

- 8 What is the sum of the real roots of the equation

$$\left(\frac{x^2}{x+1}\right)^2 - 4\left(\frac{x^2}{x+1}\right) = 5?$$

- A -4
- B 1
- C 4
- D 5
- E 6

- 9 A circle is tangent to both the  $x$ -axis and the  $y$ -axis. The circle passes through the point  $(1, 2)$ . What is the radius of the largest such circle?

- A 1
- B 2.5
- C 3
- D 5
- E  $\sqrt{5}$

10 Which of the following equations has no real solutions?

- A  $x^3 + x = 3$
- B  $\ln(x) = 2 - x$
- C  $e^x = 2 - x^2$
- D  $\sin(x) = x - 1$
- E  $3^x + 3^{-x} = \frac{3}{2}$

11 A function  $f(x)$  has derivative  $f'(x) = 12x^2 - 12x$ .

What is the value of the definite integral

$$\int_0^1 f(x) dx$$

- A -2
- B -1
- C 0
- D 1
- E 2
- F Cannot be determined from the information given

12 A geometric sequence  $T_n$  is defined for  $n = 1, 2, 3, \dots$  by

$$T_n = 8(4^{n-1})$$

A second sequence  $U_n$  is defined by  $U_n = \log_2(T_n)$ .

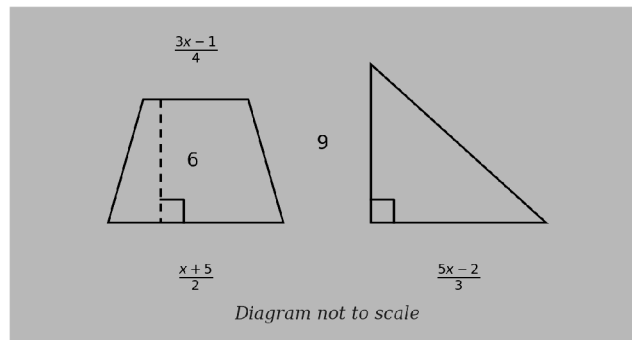
Find the value of the positive integer  $N$  for which the sum of the first  $N$  terms of the sequence  $U_n$  is 168.

- A 8
- B 11
- C 12
- D 14
- E 16
- F 18

- 13** The set of values of  $x$  for which  $\frac{6}{x-1} \geq x$  is given by:
- A**  $1 < x \leq 3$
  - B**  $-2 \leq x \leq 3$
  - C**  $x \leq -2$  or  $1 < x \leq 3$
  - D**  $-2 \leq x < 1$  or  $x \geq 3$
  - E**  $x \leq -2$  or  $x \geq 3$
- 14** A particle undergoes a displacement of 12 units on a bearing of  $060^\circ$ , followed by a displacement of  $L$  units on a bearing of  $330^\circ$ .  
The final position of the particle is due North of the starting point.  
What is the value of  $L$ ?
- A**  $4\sqrt{3}$
  - B**  $6\sqrt{3}$
  - C** 12
  - D**  $12\sqrt{3}$
  - E** 24
- 15** The polynomial  $f(x) = (x + 3)^{10}$  is expanded in powers of  $x$ .  
How many of the coefficients in this expansion are divisible by 3?
- A** 1
  - B** 9
  - C** 10
  - D** 11
  - E** 5

- 16** The diagram shows a trapezium and a right-angled triangle, with dimensions given in terms of  $x$ .  
All given lengths must be strictly positive.

Find the complete set of values of  $x$  for which the area of the trapezium is strictly greater than the area of the triangle.



- A**  $x < \frac{13}{5}$   
**B**  $x > \frac{13}{5}$   
**C**  $\frac{1}{3} < x < \frac{13}{5}$   
**D**  $\frac{2}{5} < x < \frac{13}{5}$   
**E**  $x < \frac{4}{3}$   
**F**  $\frac{1}{3} < x < \frac{4}{3}$   
**G**  $\frac{2}{5} < x < \frac{4}{3}$   
**H**  $\frac{2}{5} < x < \frac{31}{15}$
- 18** A geometric series has first term  $\ln(\sqrt{2} \cos x)$  and common ratio  $\sin^2 x$ , for an angle  $x$  in the range  $0 < x < \frac{\pi}{2}$ .

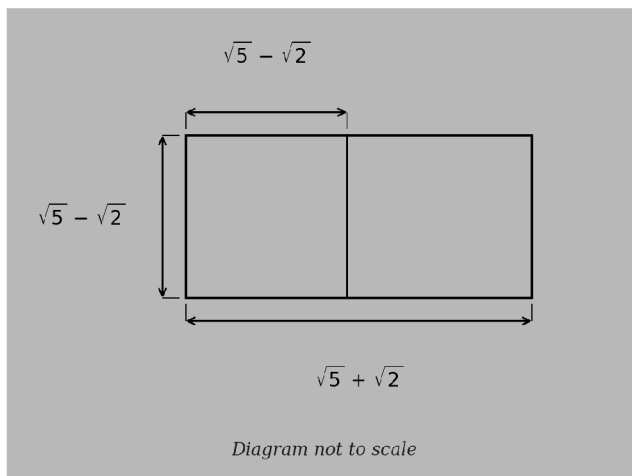
The sum to infinity of the series is 1.

What is the value of  $x$ ?

- A**  $\frac{\pi}{6}$   
**B**  $\frac{\pi}{4}$   
**C**  $\frac{\pi}{3}$   
**D**  $\frac{\pi}{2}$   
**E**  $\arccos\left(\frac{1}{\sqrt{2}}\right)$

- 19 The diagram shows a rectangle of length  $\sqrt{5} + \sqrt{2}$  and width  $\sqrt{5} - \sqrt{2}$ . A square of side length  $\sqrt{5} - \sqrt{2}$  is removed from one end, leaving the shaded region.

What is the area of the shaded region?



- A  $2\sqrt{10} - 4$   
 B  $2\sqrt{10} + 4$   
 C  $\sqrt{10} - 4$   
 D  $4 - 2\sqrt{10}$   
 E  $7 - 2\sqrt{10}$
- 20 The curve with equation  $y = x^3 - 3x^2$  and the line with equation  $y = 9x + k$  intersect at exactly three distinct points.  
 Which of the following describes all possible values of the constant  $k$ ?
- A  $-27 < k < 5$   
 B  $-27 \leq k \leq 5$   
 C  $-5 < k < 27$   
 D  $-5 \leq k \leq 27$   
 E  $k < -27$  or  $k > 5$
- 21 Let  $x$  and  $y$  be positive real numbers such that the expression below is defined.

$$\frac{2 - \log_x(y)}{(\log_x(y))^2 + \log_x(y) - 6} = \log_k(x)$$

Find  $k$  in terms of  $x$  and  $y$ .

- A  $\frac{1}{yx^3}$   
 B  $yx^3$   
 C  $\frac{x^3}{y}$   
 D  $\frac{y}{x^3}$

- 22 A function  $f(x)$  is defined by

$$f(x) = \frac{2x^3 + 2x^2 - 12x}{x^2 + x - 6}$$

What is the coefficient of  $x^3$  in the expansion of  $(1 - f(x))^5$ ?

- A** 80  
**B** -40  
**C** -80  
**D** -10  
**E** 10  
**F** -480
- 23 A geometric progression  $x_1, x_2, x_3, \dots$  consists of positive real numbers.  
 Let  $S$  be the sum  $S = \sum_{n=1}^{15} \log_b(x_n)$ , where  $b > 1$ .  
 Which one of the following pieces of information is sufficient to determine the value of  $S$ ?
- A** The value of  $x_7 + x_9$  is known.  
**B** The value of  $x_4 x_{12}$  is known.  
**C** The value of  $\sum_{n=1}^{15} x_n$  is known.  
**D** The value of  $x_1 x_{16}$  is known.  
**E** The value of  $\prod_{n=1}^{14} x_n$  is known.
- 24 A triangle has vertices at  $(0, 0)$ ,  $(x, 0)$ , and  $(x, f(x))$  for  $x > 0$ . The area of the triangle,  $A$ , is a function of  $x$ .  
 The rate of change of the area with respect to  $x$  is given by

$$\frac{dA}{dx} = 2 \sin^2(x)$$

Given that  $A = 0$  when  $x = 0$ , what is the value of  $f\left(\frac{\pi}{4}\right)$ ?

- A**  $2 - \frac{4}{\pi}$   
**B**  $\frac{8}{\pi}$   
**C**  $1 - \frac{2}{\pi}$   
**D**  $2 + \frac{4}{\pi}$   
**E**  $1 + \frac{2}{\pi}$   
**F**  $\frac{16}{\pi}$

- 25 Which one of the following is a simplification of  $5 - \frac{9 - 4x^2}{2x^2 - 3x}$  for all valid values of  $x$ ?

- A  $3 - \frac{3}{x}$
- B  $3 + \frac{3}{x}$
- C  $5 - \frac{3}{x}$
- D  $5 + \frac{3}{x}$
- E  $7 - \frac{3}{x}$
- F  $7 + \frac{3}{x}$

- 26 Find the exact value of the sum

$$\sum_{k=1}^{10} \log_3 \left( \frac{\sqrt{3}}{27^{k-1}} \right)$$

- A -95
- B -85
- C -75
- D -70
- E -55
- F 195

- 27 A function  $f$  is defined for  $x > 0$  by

$$f(x) = \frac{(2x^2)^{-3}}{x^{-10}}$$

What is the value of  $f'(1)$ ?

- A  $\frac{1}{2}$
- B -32
- C -2
- D  $\frac{1}{8}$
- E 8
- F  $\frac{15}{2}$

## Physics

- 1 A magnet rests on a top-pan balance. A straight horizontal wire of length  $4.0\text{ cm}$  is held between the magnet's poles and carries current  $I$ . The balance reading increases linearly from  $500.0\text{ g}$  at  $I = 0$  to  $504.0\text{ g}$  at  $I = 2.0\text{ A}$ . What is the magnitude of the magnetic field strength between the poles? Take  $g = 10\text{ N kg}^{-1}$ .

- A 0.0050 T  
B 0.050 T  
C 0.50 T  
D 2.0 T  
E 5.0 T  
F 50 T

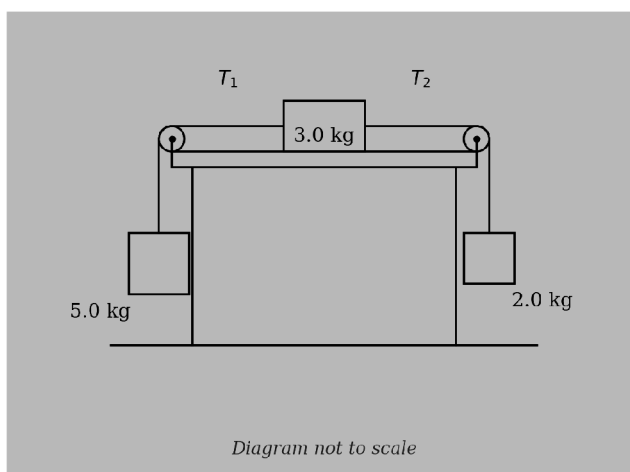
- 2 A block of mass  $3.0\text{ kg}$  rests on a smooth, horizontal table. It is connected by two light, inextensible strings passing over smooth, frictionless pulleys at opposite ends of the table to two hanging masses of  $5.0\text{ kg}$  and  $2.0\text{ kg}$ , as shown in the diagram.

The system is released from rest.

What is the tension  $T_1$  in the string attached to the  $5.0\text{ kg}$  mass and the tension  $T_2$  in the string attached to the  $2.0\text{ kg}$  mass while the blocks are moving?

[Take the acceleration due to gravity to be  $g = 10\text{ m s}^{-2}$ .]

|   | $T_1 / \text{N}$ | $T_2 / \text{N}$ |
|---|------------------|------------------|
| A | 15               | 6                |
| B | 15               | 20               |
| C | 35               | 20               |
| D | 35               | 26               |
| E | 50               | 20               |
| F | 50               | 26               |
| G | 65               | 14               |
| H | 65               | 20               |



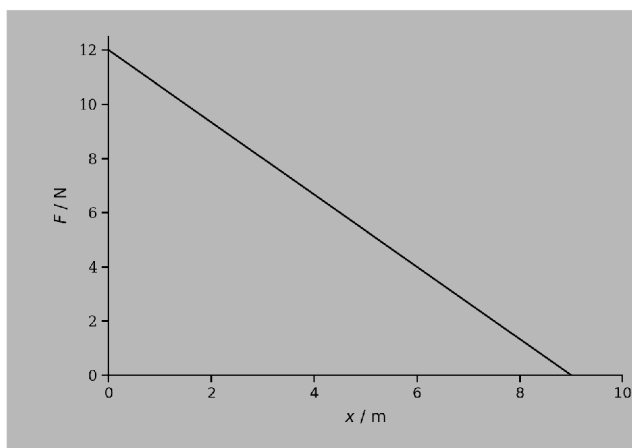
- 3 A laser beam reflects off a plane mirror  $M_1$  and then off a second plane mirror  $M_2$ . The direction of the beam incident on  $M_1$  is fixed.  $M_1$  is rotated by an angle  $\alpha$  in a clockwise direction.

To ensure the final beam reflected from  $M_2$  maintains a constant direction,  $M_2$  must be rotated by:

- A  $\alpha$  in the clockwise direction
  - B  $\alpha$  in the anticlockwise direction
  - C  $2\alpha$  in the clockwise direction
  - D  $2\alpha$  in the anticlockwise direction
  - E  $\frac{\alpha}{2}$  in the clockwise direction
- 4 A block of mass 2.0 kg is initially moving at a speed of  $4.0 \text{ m s}^{-1}$  along a smooth horizontal surface in the positive  $x$ -direction.

A single horizontal force  $F$  acts on the block in the direction of its motion. The graph shows how  $F$  varies with the displacement  $x$  of the block.

What is the speed of the block when it reaches  $x = 6.0 \text{ m}$ ?



- A  $6.0 \text{ m s}^{-1}$
- B  $6.9 \text{ m s}^{-1}$
- C  $7.2 \text{ m s}^{-1}$
- D  $8.0 \text{ m s}^{-1}$
- E  $8.4 \text{ m s}^{-1}$
- F  $9.4 \text{ m s}^{-1}$

- 5 The vibration of the  $\text{H} - \text{Cl}$  bond can be modelled by assuming the chlorine atom is stationary and the hydrogen atom oscillates with simple harmonic motion.

The frequency of the vibration is  $3.0$

$\times 10^{13}$

$\text{Hz}$  and the amplitude (maximum displacement from the equilibrium position) is  $5.0$

$\times 10^{-12}$

$\text{m}$ .

What is the average speed of the hydrogen atom over one full period of oscillation?

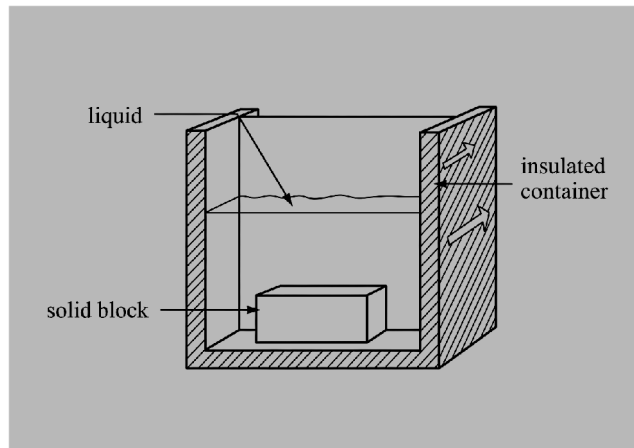
- A  $1.5$   
 $\times 10^2$   
 $\text{ms}^{-1}$
- B  $3.0$   
 $\times 10^2$   
 $\text{ms}^{-1}$
- C  $6.0$   
 $\times 10^2$   
 $\text{ms}^{-1}$
- D  $9.4$   
 $\times 10^2$   
 $\text{ms}^{-1}$
- E  $3.0$   
 $\times 10^8$   
 $\text{ms}^{-1}$

- 6 A hot solid block is placed into a cold liquid in an insulated container.

The mass of the liquid is twice the mass of the block. The specific heat capacity of the liquid is three times the specific heat capacity of the block.

The temperature of the block decreases by  $\Delta T$ .

Assuming no thermal energy is transferred to the surroundings, what is the temperature increase of the liquid?



concept image

- A  $\frac{\Delta T}{6}$
- B  $\frac{\Delta T}{5}$
- C  $\frac{\Delta T}{3}$
- D  $\frac{\Delta T}{2}$
- E  $5\Delta T$
- F  $6\Delta T$

- 7 The activity of a radioactive source is measured. Its activity after 1 hour is 8 times its activity after 4 hours.

Its activity after 2 hours is 120 Bq.

What was the initial activity of the source in Bq?

- A 30
- B 120
- C 240
- D 480
- E 960
- F 1920

- 8** A sealed syringe with a freely moving, frictionless piston contains a fixed mass of an ideal gas. At a depth of 30 m below the surface of a lake, the volume of the gas in the syringe is  $80 \text{ cm}^3$ . When the syringe is moved to a depth of 10 m below the surface, the volume of the gas increases to  $120 \text{ cm}^3$ .

The temperature of the water is uniform throughout the lake, and the gas remains in thermal equilibrium with the water.

What is the volume of the gas in the syringe at the surface of the lake?

- A**  $135 \text{ cm}^3$
  - B**  $140 \text{ cm}^3$
  - C**  $150 \text{ cm}^3$
  - D**  $160 \text{ cm}^3$
  - E**  $180 \text{ cm}^3$
  - F**  $200 \text{ cm}^3$
  - G**  $240 \text{ cm}^3$
  - H**  $360 \text{ cm}^3$
- 9** Two cylindrical rods, X and Y, are used as thermal conductors between two reservoirs maintained at a constant temperature difference. Rod Y has 3 times the thermal conductivity, twice the length, and twice the diameter of rod X.

In the steady state, the rate of heat flow through rod X is  $P_X$  and the rate of heat flow through rod Y is  $P_Y$ .

What is the value of the ratio  $\frac{P_Y}{P_X}$ ?

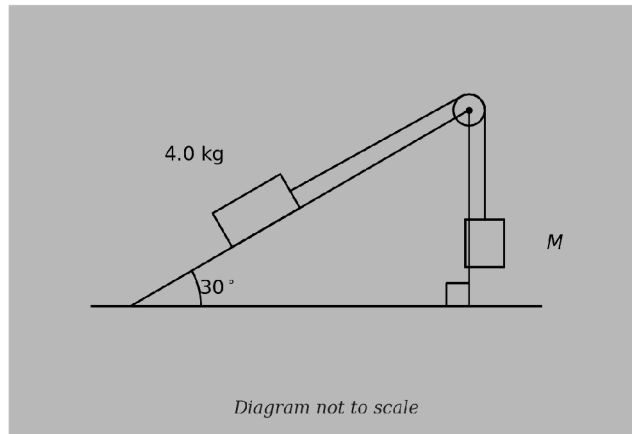
- A** 2
- B** 3
- C** 6
- D** 12
- E** 24

- 10** A block of mass 4.0 kg is placed on a rough plane inclined at an angle of  $30^\circ$  to the horizontal.

The coefficient of static friction between the block and the plane is  $\mu = \frac{\sqrt{3}}{6}$ .

The block is attached to a light, inextensible string that passes over a smooth, frictionless pulley fixed at the top of the incline and supports a hanging mass  $M$ , as shown in the diagram.

What is the range of possible values for the hanging mass  $M$  such that the system remains in static equilibrium?



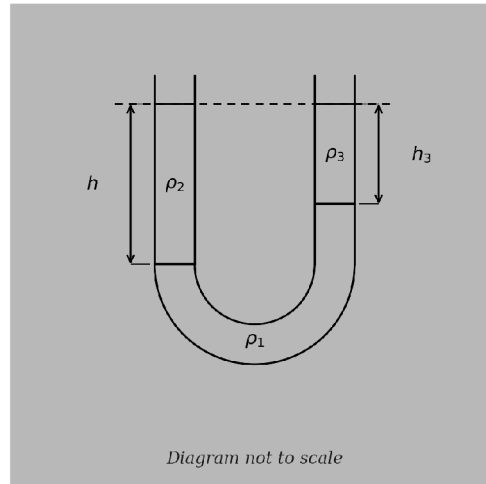
- A**  $1.0 \text{ kg} \leq M \leq 3.0 \text{ kg}$
- B**  $1.5 \text{ kg} \leq M \leq 2.5 \text{ kg}$
- C**  $1.0 \text{ kg} \leq M \leq 2.0 \text{ kg}$
- D**  $2.0 \text{ kg} \leq M \leq 3.0 \text{ kg}$
- E**  $2.0 \text{ kg} \leq M \leq 4.0 \text{ kg}$

- 11 A U-tube of uniform cross-sectional area, open to the atmosphere at both ends, initially contains a liquid of density  $\rho_1$ .

A second immiscible liquid of density  $\rho_2$  is poured into the left arm, forming a column of height  $h$ .

A third immiscible liquid of density  $\rho_3$  is then poured into the right arm until the top liquid surfaces in both arms are at the exact same horizontal level, as shown in the diagram.

Given that  $\rho_1 > \rho_2 > \rho_3$ , which expression gives the height  $h_3$  of the column of the third liquid?



- A  $\left(\frac{\rho_2}{\rho_3}\right)h$
- B  $\left(\frac{\rho_3}{\rho_2}\right)h$
- C  $\left(\frac{\rho_1 - \rho_2}{\rho_1 - \rho_3}\right)h$
- D  $\left(\frac{\rho_1 - \rho_3}{\rho_1 - \rho_2}\right)h$
- E  $\left(\frac{\rho_2 - \rho_3}{\rho_1 - \rho_3}\right)h$
- F  $\left(\frac{\rho_1 - \rho_2}{\rho_2 - \rho_3}\right)h$
- G  $\left(\frac{\rho_1 + \rho_2}{\rho_1 + \rho_3}\right)h$
- H  $\left(\frac{\rho_1 - \rho_2}{\rho_1 + \rho_3}\right)h$

- 12** Plane water waves travel along a straight canal. At a stationary post, wave crests are observed to pass every 2.0 s. A boat travels along the canal in the same direction as the waves with a constant speed of 4.0 m/s. An observer on the boat records that wave crests overtake the boat every 6.0 s.

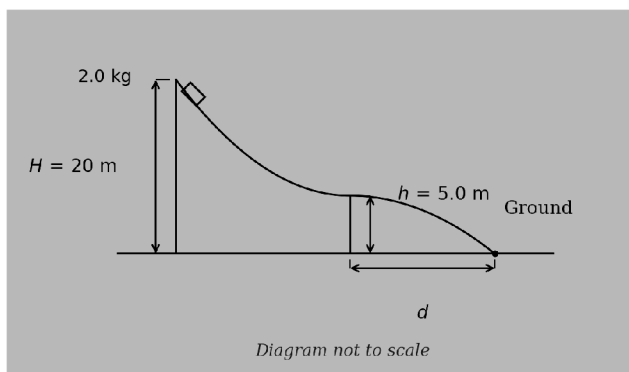
What is the speed of the waves?

- A** 2.0 m/s  
**B** 3.0 m/s  
**C** 4.0 m/s  
**D** 6.0 m/s  
**E** 12.0 m/s
- 13** A small block of mass 2.0 kg is released from rest at the top of a smooth, curved ramp at a height  $H = 20$  m above horizontal ground. The end of the ramp is horizontal and is at a height  $h = 5.0$  m above the ground, where the block leaves the ramp.

Air resistance is negligible.

The gravitational field strength is  $g = 10 \text{ N kg}^{-1}$ .

What is the horizontal distance  $d$  travelled by the block between leaving the ramp and landing on the ground?



- A**  $5\sqrt{3} \text{ m}$   
**B** 10 m  
**C**  $10\sqrt{3} \text{ m}$   
**D** 15 m  
**E** 20 m  
**F**  $20\sqrt{3} \text{ m}$

- 14** Three identical resistors are connected in a circuit with a DC power supply. One resistor is connected in series with a parallel combination of the other two.  
What fraction of the total power supplied by the source is dissipated by the parallel combination?
- A**  $\frac{1}{9}$   
**B**  $\frac{1}{6}$   
**C**  $\frac{1}{4}$   
**D**  $\frac{1}{3}$   
**E**  $\frac{1}{2}$   
**F**  $\frac{2}{3}$
- 15** An electric pump lifts oil of density  $800 \text{ kg m}^{-3}$  through a vertical height of 5.0 m at a constant volumetric flow rate of  $2.5 \times 10^{-3} \text{ m}^3 \text{ s}^{-1}$ . The pump system has an overall efficiency of 50% and is connected to a 50 V DC power supply.  
Assuming  $g = 10 \text{ m s}^{-2}$ , what is the current drawn from the supply?
- A**  $1.0 \text{ extA}$   
**B**  $2.0 \text{ extA}$   
**C**  $2.5 \text{ extA}$   
**D**  $4.0 \text{ extA}$   
**E**  $5.0 \text{ extA}$
- 16** A continuous wave travels from medium P, where its speed is  $v$ , into medium Q, where its speed is  $0.8v$ .  
Which of the following gives the fractional change in the wavelength of the wave?
- A**  $-\frac{1}{4}$   
**B**  $-\frac{1}{5}$   
**C** 0  
**D**  $\frac{1}{5}$   
**E**  $\frac{1}{4}$

- 17** A solid cuboid is made from a material of density  $2000 \text{ kg m}^{-3}$ . Its side lengths are in the ratio 1 : 2 : 4. The cuboid is placed on a horizontal surface to exert the maximum possible pressure. This pressure is 8.0 kPa.

What is the mass of the cuboid?

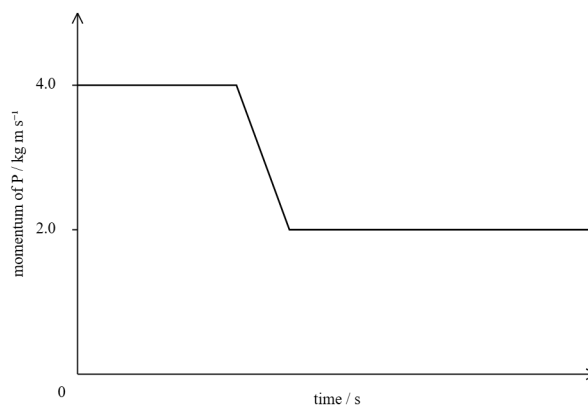
(The gravitational field strength  $g$  is  $10 \text{ N kg}^{-1}$ )

- A** 2.0 kg
  - B** 16 kg
  - C** 128 kg
  - D** 160 kg
  - E** 1024 kg
  - F** 16000 kg
- 18** A particle is projected from a point  $O$  with speed  $u$ . The particle passes through a target point  $P$  located at a horizontal distance  $d$  and vertical height  $h$  from  $O$ .

Which of the following expressions gives the minimum value of  $u$  required for the particle to reach  $P$ ?

- A**  $\sqrt{g}$
- B**  $\sqrt{2}$
- C**  $\sqrt{g}$
- D**  $\sqrt{2}$
- E**  $\sqrt{g}$
- F**  $\sqrt{g}$

- 19** An object P collides with a stationary object Q on a straight, frictionless track. The graph shows the momentum of object P as a function of time.



What is the momentum of object Q after the collision?

- A**  $-4.0 \text{ kg m s}^{-1}$
  - B**  $-2.0 \text{ kg m s}^{-1}$
  - C**  $0 \text{ kg m s}^{-1}$
  - D**  $2.0 \text{ kg m s}^{-1}$
  - E**  $4.0 \text{ kg m s}^{-1}$
  - F**  $6.0 \text{ kg m s}^{-1}$
- 20** An electromagnet is placed on a top-pan balance and produces a uniform horizontal magnetic field of strength  $0.50 \text{ T}$ .  
A horizontal beam of protons is fired through the field, perpendicular to the field lines. The experiment is then repeated with a beam of electrons at the same speed, travelling in the same direction.  
At any instant, there are  $1.0 \times 10^{10}$  particles within the magnetic field.  
The difference between the two balance readings is  $1.6 \text{ g}$ .

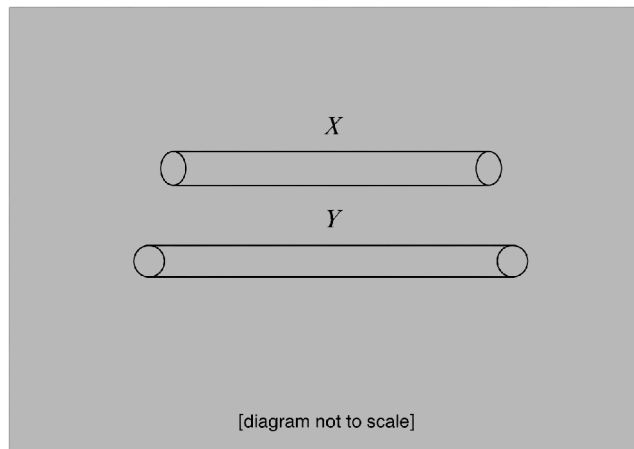
What is the speed of the particles?

(elementary charge,  $e = 1.6 \times 10^{-19} \text{ C}$ ; gravitational field strength,  $g = 10 \text{ N kg}^{-1}$ )

- A**  $1.0 \times 10^5 \text{ m s}^{-1}$
- B**  $2.0 \times 10^7 \text{ m s}^{-1}$
- C**  $1.0 \times 10^7 \text{ m s}^{-1}$
- D**  $4.0 \times 10^7 \text{ m s}^{-1}$
- E**  $1.0 \times 10^{10} \text{ m s}^{-1}$

- 21** Wire X and wire Y are made of the same material. Wire Y has twice the length and half the diameter of wire X.

What is the ratio of their electrical resistances,  $\frac{R_Y}{R_X}$ ?



concept image

- A**  $\frac{1}{8}$   
**B**  $\frac{1}{4}$   
**C**  $\frac{1}{2}$   
**D** 2  
**E** 4  
**F** 8
- 22** An ideal gas bubble of volume  $V$  is released at a depth  $3H$  below the surface of a liquid. It rises to a depth  $H$ .  
 The atmospheric pressure at the surface is equal to the pressure exerted by a column of the liquid of height  $H$ .  
 Assuming the temperature of the gas remains constant, what is the volume of the bubble at depth  $H$ ?
- A**  $\frac{V}{3}$   
**B**  $\frac{V}{2}$   
**C**  $2V$   
**D**  $3V$   
**E**  $4V$

- 23** A laser beam from a fixed source is incident on a plane mirror  $M_1$ . The reflected beam then strikes a second plane mirror  $M_2$ .  $M_1$  rotates with angular speed  $\omega$  and  $M_2$  rotates with angular speed  $4\omega$ . Both mirrors rotate in the same direction about parallel axes perpendicular to the plane of incidence.

What is the angular speed of the final beam reflected from  $M_2$ ?

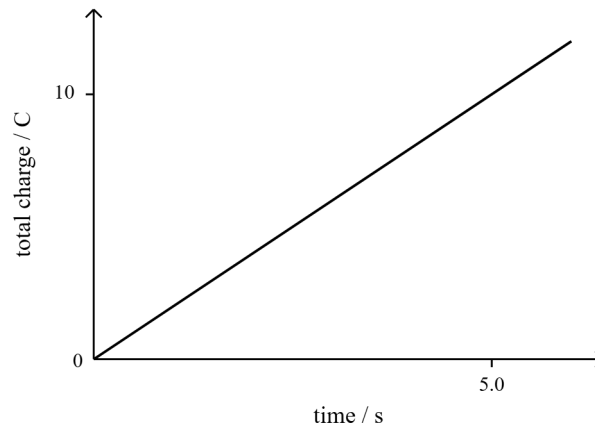
- A**  $3\omega$
  - B**  $5\omega$
  - C**  $6\omega$
  - D**  $8\omega$
  - E**  $10\omega$
- 24** A solid cuboid of mass  $1.0 \text{ kg}$  is placed on a horizontal surface. Depending on which face it rests on, it can exert one of three possible pressures:  $0.40 \text{ N cm}^{-2}$ ,  $0.50 \text{ N cm}^{-2}$ , or  $2.0 \text{ N cm}^{-2}$ .

What is the density of the cuboid's material, in  $\text{g cm}^{-3}$ ?

(Take the gravitational field strength  $g$  to be  $10 \text{ N kg}^{-1}$ .)

- A** 0.05
- B** 0.40
- C** 2.0
- D** 2.5
- E** 20
- F** 20000

- 25** A resistor of resistance  $4.0\ \Omega$  is connected to a power supply. The graph shows the total charge that has passed through the resistor as a function of time.



What is the potential difference across the resistor?

- A** 0.50 V
  - B** 2.0 V
  - C** 8.0 V
  - D** 20 V
  - E** 25 V
  - F** 40 V
- 26** A charged particle of mass  $m$  is released from rest in a uniform field. The particle experiences a constant force  $F$  due to the field.

What is the kinetic energy of the particle after it has travelled a distance  $d$  parallel to the force?

- A**  $\frac{1}{2}Fd$
- B**  $2Fd$
- C**  $Fd$
- D**  $\frac{Fd}{m}$
- E**  $Fdm$

- 27** The electrical resistance  $R$  of a uniform metal wire depends on its length  $L$  and cross-sectional area  $A$ .

Which of the following proportionalities is correct?

- A**  $R \propto LA$
- B**  $R \propto \frac{A}{L}$
- C**  $R \propto L$
- D**  $R \propto \frac{1}{A}$
- E**  $R \propto \frac{L}{A}$

## Chemistry

- 1 Three liquid fractions (P, Q, and R) are extracted from a biological sample. Their physical properties are shown in the table below.

| Fraction | Density / g cm <sup>-3</sup> | Boiling point / °C | Miscibility             |
|----------|------------------------------|--------------------|-------------------------|
| P        | 0.79                         | 56                 | Miscible with Q         |
| Q        | 1.00                         | 100                | Miscible with P         |
| R        | 1.49                         | 61                 | Immiscible with P and Q |

The fractions are mixed in a separating funnel and allowed to settle, forming two layers.

Which procedure describes the correct method to obtain a pure sample of fraction P?

- A Collect the upper layer and heat to 56°C  
 B Collect the upper layer and heat to 100°C  
 C Collect the lower layer and heat to 56°C  
 D Collect the lower layer and heat to 61°C  
 E Collect the upper layer and heat to 61°C
- 2 A student conducts two experiments to heat liquids using the energy released from the combustion of a fixed amount of ethanol. Assume that there is no heat loss and that the amount of energy transferred to the liquid is the same in both experiments.

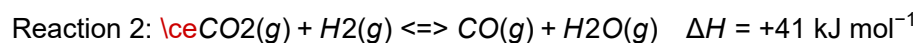
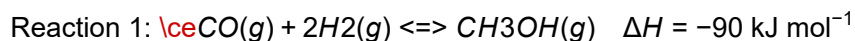
In Experiment 1, mass  $m$  of liquid X (specific heat capacity  $c$ ) is heated, and the temperature rises by  $\Delta T$ .

In Experiment 2, mass  $3m$  of liquid Y (specific heat capacity  $\frac{1}{2}c$ ) is heated.

Which expression gives the temperature rise in Experiment 2?

- A  $\frac{1}{6}\Delta T$   
 B  $\frac{1}{3}\Delta T$   
 C  $\frac{2}{3}\Delta T$   
 D  $\frac{3}{2}\Delta T$   
 E  $6\Delta T$

- 3 Two reversible gas-phase reactions are carried out in separate closed vessels:



The following actions could be applied independently to each reaction system:

- 1 decreasing the volume of the reaction container at constant temperature
- 2 increasing the temperature of the reaction container at constant volume
- 3 adding an appropriate catalyst at constant temperature and volume

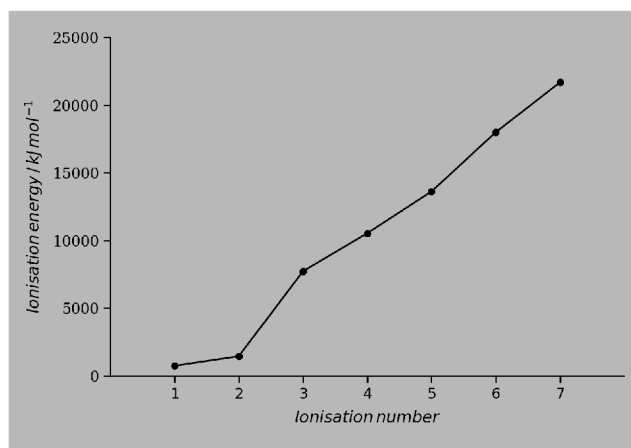
Assuming all gases behave ideally, which of these actions will increase the initial rate of the forward reaction for **both** reactions, **and** increase the equilibrium yield of products for Reaction 1, while having **no effect** on the equilibrium yield of products for Reaction 2?

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

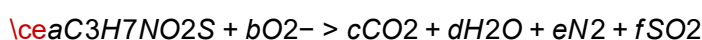
- 4 The graph shows the first seven successive ionisation energies of an element  $\text{X}$ , which belongs to Period 3 of the Periodic Table.

Which row in the table correctly identifies the formula of the simple oxide of  $\text{X}$ , the structure and bonding present in this oxide, and its acid-base character?

|   | formula of oxide       | structure and bonding | acid-base character |
|---|------------------------|-----------------------|---------------------|
| A | $\text{X}_2\text{O}$   | giant ionic           | basic               |
| B | $\text{X}_2\text{O}$   | simple covalent       | acidic              |
| C | $\text{XO}$            | giant ionic           | basic               |
| D | $\text{XO}$            | simple covalent       | acidic              |
| E | $\text{XO}_2$          | giant covalent        | acidic              |
| F | $\text{XO}_2$          | simple covalent       | acidic              |
| G | $\text{X}_2\text{O}_3$ | giant ionic           | basic               |
| H | $\text{X}_2\text{O}_3$ | giant ionic           | amphoteric          |



- 5 The complete combustion of the amino acid cysteine,  $\text{C}_3\text{H}_7\text{NO}_2\text{S}$ , produces carbon dioxide, water, nitrogen gas, and sulfur dioxide according to the equation:

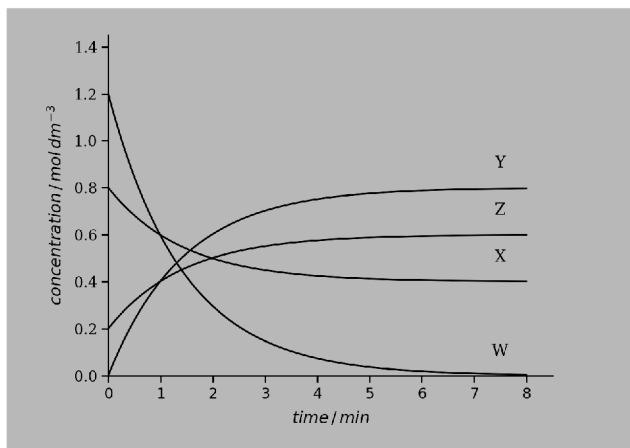


What is the value of  $b$  when the equation is balanced using the lowest whole-number coefficients?

- A 9
- B 11
- C 15
- D 17
- E 19
- F 23
- G 38

- 6 The graph shows how the concentrations of four substances,  $W$ ,  $X$ ,  $Y$ , and  $Z$ , change over time during a reaction in a closed vessel at constant temperature.

Which of the following is the balanced equation for this reaction?

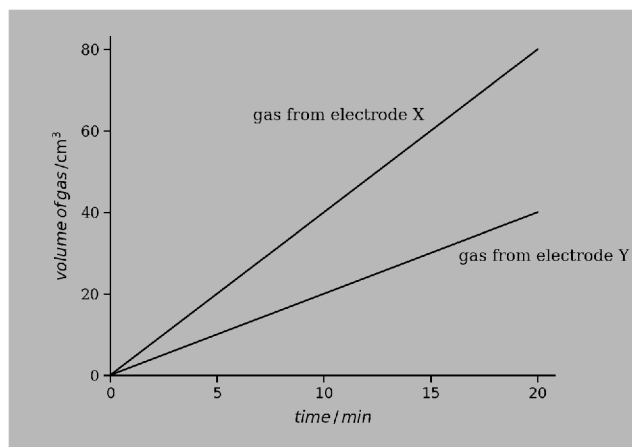


- A  $3W + X \rightarrow 2Y + Z$   
 B  $3W + X \rightarrow 4Y + 3Z$   
 C  $3W + 2X \rightarrow 2Y + Z$   
 D  $3W + 2X \rightarrow 2Y + 3Z$   
 E  $6W + 4X \rightarrow 4Y + 3Z$   
 F  $3W \rightarrow X + 2Y + Z$

- 7 The graph shows the volume of gas collected at each of two inert electrodes, X and Y, during the electrolysis of dilute aqueous sulfuric acid,  $\text{H}_2\text{SO}_4(\text{aq})$ , at constant current, temperature, and pressure.

Which of the following statements is/are correct?

1. Electrode X is connected to the negative terminal of the power supply.
2. In the external circuit, electrons flow from electrode X to electrode Y.
3. For every 1 mol of gas collected at electrode Y, 4 mol of electrons pass through the external circuit.



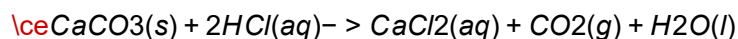
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 8 The following information about four non-metallic elements labelled W, X, Y and Z is given:
- Elements W and Y displace bromine from an aqueous solution of potassium bromide,  $\text{KBr(aq)}$ .
  - Elements X and Z cannot displace bromine from an aqueous solution of potassium bromide,  $\text{KBr(aq)}$ .
  - Element Y reacts explosively with hydrogen gas in the dark, whereas element W requires ultraviolet light to react with hydrogen gas.
  - Element Z reacts with an aqueous solution of the sodium salt  $\text{NaX}$  to displace element X.

What is the order of reactivity (oxidising ability) of these four elements, starting with the most reactive?

- A W, Y, X, Z
  - B W, Y, Z, X
  - C X, Z, W, Y
  - D X, Z, Y, W
  - E Y, W, X, Z
  - F Y, W, Z, X
  - G Z, X, W, Y
  - H Z, X, Y, W
- 9 Element X has an atomic number of 15. It reacts with hydrogen to form a simple molecular hydride.
- Which option gives the correct formula of the hydride and the shape of the molecule?
- A  $\text{XH}_2$ , linear
  - B  $\text{XH}_2$ , bent (V-shaped)
  - C  $\text{XH}_3$ , trigonal planar
  - D  $\text{XH}_3$ , trigonal pyramidal
  - E  $\text{XH}_4$ , tetrahedral

10 Calcium carbonate reacts with hydrochloric acid according to the equation:



In a baseline experiment (represented by curve **X** on the graph), 2.00 g of large marble chips ( $\text{CaCO}_3$ ) reacts with 50 cm<sup>3</sup> of 1.0 mol dm<sup>-3</sup>  $\text{HCl}(\text{aq})$  at 25 °C.

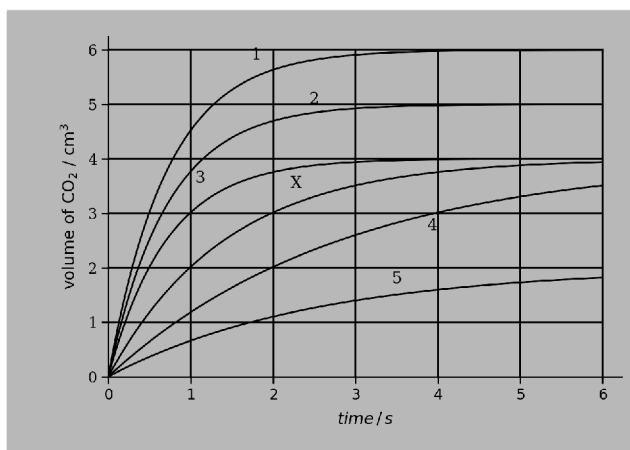
Two further experiments are carried out at 25 °C:

- **Experiment 1:** 3.00 g of powdered  $\text{CaCO}_3$  is reacted with 50 cm<sup>3</sup> of 1.0 mol dm<sup>-3</sup>  $\text{HCl}(\text{aq})$
- **Experiment 2:** 2.00 g of large marble chips ( $\text{CaCO}_3$ ) is reacted with 100 cm<sup>3</sup> of 0.40 mol dm<sup>-3</sup>  $\text{HCl}(\text{aq})$

( $M_r$  value:  $\text{CaCO}_3 = 100$ )

Which curves show how the volume of  $\text{CO}_2$  evolved changes with time in Experiment 1 and Experiment 2?

|          | Experiment 1 | Experiment 2 |
|----------|--------------|--------------|
| <b>A</b> | curve 1      | curve 4      |
| <b>B</b> | curve 1      | curve 5      |
| <b>C</b> | curve 2      | curve 4      |
| <b>D</b> | curve 2      | curve 5      |
| <b>E</b> | curve 3      | curve 4      |
| <b>F</b> | curve 3      | curve 5      |

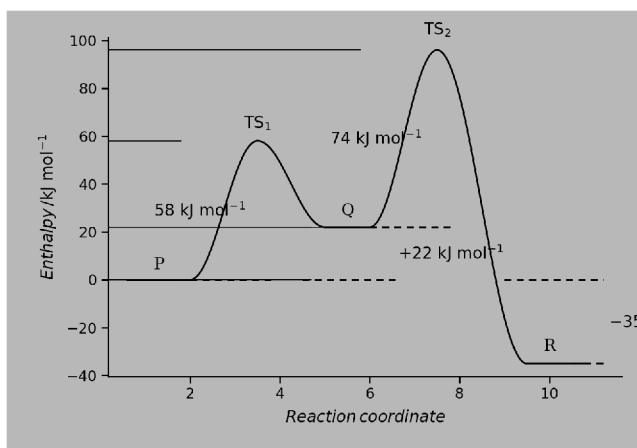


- 11 The reaction energy profile for a two-step gas-phase reaction  $P(g) \rightarrow Q(g) \rightarrow R(g)$  is shown in the diagram below.

From the information provided in the energy profile:

- The activation energy for the forward step  $P \rightarrow Q$  is  $58 \text{ kJ mol}^{-1}$ .
- The intermediate  $Q$  has an enthalpy  $22 \text{ kJ mol}^{-1}$  higher than reactant  $P$ .
- The activation energy for the forward step  $Q \rightarrow R$  is  $74 \text{ kJ mol}^{-1}$ .
- The overall enthalpy change for  $P \rightarrow R$  is  $\Delta H = -35 \text{ kJ mol}^{-1}$ .

What are the activation energy for the conversion of intermediate  $Q$  to reactant  $P$  ( $E_a(Q \rightarrow P)$ ) and the activation energy for the conversion of product  $R$  to intermediate  $Q$  ( $E_a(R \rightarrow Q)$ )?



- A  $E_a(Q \rightarrow P) = 36 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 61 \text{ kJ mol}^{-1}$   
 B  $E_a(Q \rightarrow P) = 36 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 109 \text{ kJ mol}^{-1}$   
 C  $E_a(Q \rightarrow P) = 36 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 131 \text{ kJ mol}^{-1}$   
 D  $E_a(Q \rightarrow P) = 58 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 109 \text{ kJ mol}^{-1}$   
 E  $E_a(Q \rightarrow P) = 58 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 131 \text{ kJ mol}^{-1}$   
 F  $E_a(Q \rightarrow P) = 80 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 39 \text{ kJ mol}^{-1}$   
 G  $E_a(Q \rightarrow P) = 80 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 109 \text{ kJ mol}^{-1}$   
 H  $E_a(Q \rightarrow P) = 80 \text{ kJ mol}^{-1}$ ,  $E_a(R \rightarrow Q) = 131 \text{ kJ mol}^{-1}$

- 12 A mixture containing three volatile organic compounds,  $P$ ,  $Q$ , and  $R$ , is analysed using gas chromatography. The resulting chromatogram is shown below.

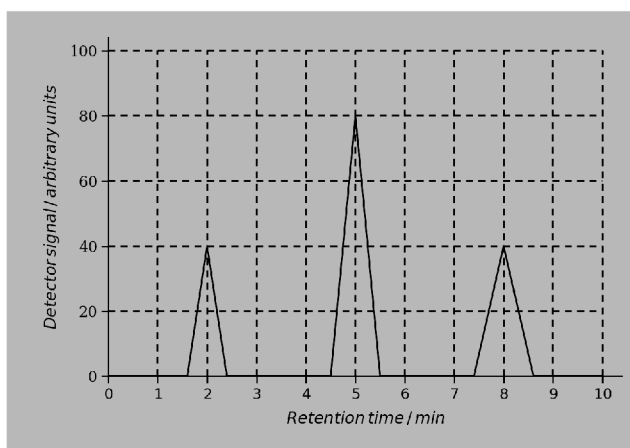
Standard samples of the pure compounds analysed under identical conditions gave the following retention times:

- Compound  $P$ : 2.0 min
- Compound  $Q$ : 5.0 min
- Compound  $R$ : 8.0 min

Assume that the detector response per unit mass is identical for all three compounds, and that each peak can be approximated as a triangle (area =  $\frac{1}{2} \times \text{base width} \times \text{height}$ ).

Which of the following statements is/are correct?

- 1 Compound  $Q$  makes up 50% of the total mass of the mixture.
- 2 Under these conditions, compound  $R$  has a stronger affinity for the stationary phase than compound  $P$ .
- 3 Compound  $P$  makes up 25% of the total mass of the mixture.



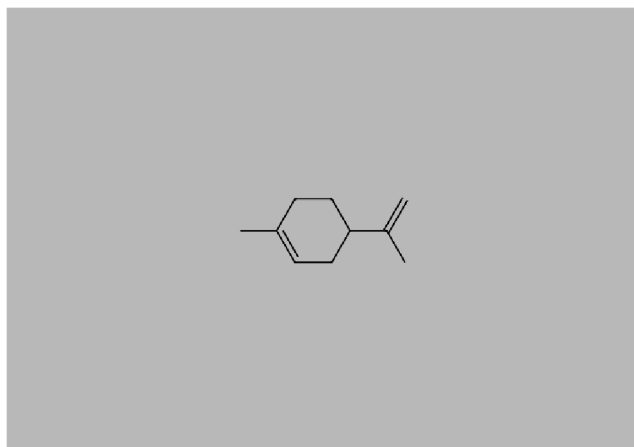
- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

**13** The skeletal structure of limonene is shown below:

Limonene undergoes complete catalytic hydrogenation to form a fully saturated hydrocarbon.

What is the minimum volume of hydrogen gas, measured at room temperature and pressure (RTP), required to react completely with 1.60 cm<sup>3</sup> of limonene?

( $M_r$  of limonene = 136; density of limonene = 0.85 g cm<sup>-3</sup>; molar volume of a gas at RTP = 24.0 dm<sup>3</sup> mol<sup>-1</sup>)



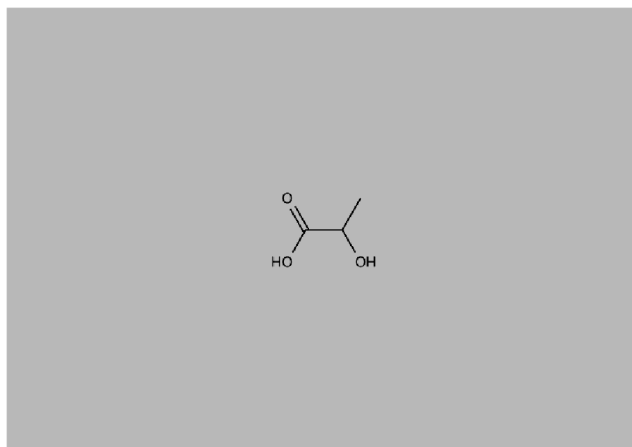
- A** 0.024 dm<sup>3</sup>
- B** 0.048 dm<sup>3</sup>
- C** 0.24 dm<sup>3</sup>
- D** 0.48 dm<sup>3</sup>
- E** 0.72 dm<sup>3</sup>
- F** 0.96 dm<sup>3</sup>
- G** 2.4 dm<sup>3</sup>
- H** 4.8 dm<sup>3</sup>

- 14** A 0.45 g sample of the organic compound shown below reacts completely with an excess of sodium metal at room temperature.

Any gas produced is collected and its volume is measured at room temperature and pressure (r.t.p.).

Assuming 1.0 mol of gas occupies 24.0 dm<sup>3</sup> at r.t.p., what volume of gas is collected?

( $A_r$  values:  $H = 1.0$ ;  $C = 12.0$ ;  $O = 16.0$ )



- A** 0.12 cm<sup>3</sup>
- B** 60 cm<sup>3</sup>
- C** 120 cm<sup>3</sup>
- D** 180 cm<sup>3</sup>
- E** 240 cm<sup>3</sup>
- F** 360 cm<sup>3</sup>
- G** 600 cm<sup>3</sup>

- 15** Isolated mitochondria are suspended in a buffer containing pyruvate as the sole respiratory substrate. The table below shows the net yield of reduced coenzymes and  $\text{ceCO}_2$  for the relevant stages of respiration.

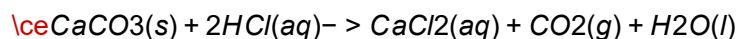
| Stage                        | Reduced NAD | Reduced FAD | $\text{ceCO}_2$ |
|------------------------------|-------------|-------------|-----------------|
| Link reaction (per pyruvate) | 1           | 0           | 1               |
| Krebs cycle (per acetyl CoA) | 3           | 1           | 2               |

In the electron transport chain, the oxidation of one molecule of reduced NAD or reduced FAD releases two electrons. Four electrons are required to reduce one molecule of oxygen ( $\text{ceO}_2$ ).

If the mitochondrial suspension produces 3.0 moles of  $\text{ceCO}_2$ , how many moles of  $\text{ceO}_2$  are consumed?

- A** 2.0  
**B** 2.5  
**C** 3.0  
**D** 5.0
- 16** Propan-2-ol is warmed with acidified potassium dichromate(VI) solution. Which of the following statements about this reaction is/are correct?
- 1 The reaction is classified as a substitution reaction.  
 2 The organic product gives a positive test with Tollens' reagent.  
 3 The organic product is a ketone.
- A** none of them  
**B** 1 only  
**C** 2 only  
**D** 3 only  
**E** 1 and 2 only  
**F** 1 and 3 only  
**G** 2 and 3 only  
**H** 1, 2 and 3

- 17 An experiment investigates the reaction between solid marble chips (calcium carbonate,  $\text{CaCO}_3$ ) and an excess of dilute hydrochloric acid:



In a second experiment, the same mass of  $\text{CaCO}_3$  is used, but the solid is ground into a fine powder. All other conditions, including the volume, concentration, and initial temperature of the hydrochloric acid, are kept exactly the same.

How do the frequency of collisions between reactant particles, the activation energy of the reaction, and the initial rate of reaction in the second experiment compare to those in the first experiment?

|   | Frequency of collisions between reactant particles | Activation energy | Initial rate of reaction |
|---|----------------------------------------------------|-------------------|--------------------------|
| A | decreases                                          | decreases         | decreases                |
| B | decreases                                          | unchanged         | decreases                |
| C | unchanged                                          | decreases         | increases                |
| D | unchanged                                          | unchanged         | increases                |
| E | increases                                          | decreases         | increases                |
| F | increases                                          | increases         | increases                |
| G | increases                                          | unchanged         | increases                |
| H | increases                                          | unchanged         | unchanged                |

- 18 Concentrated aqueous solutions of the following three compounds are electrolysed separately using inert electrodes: 1.

$\text{NaCl}$  2.

$\text{HBr}$  3.

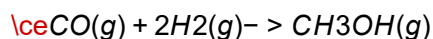
$\text{CuCl}_2$

Which of the solutions yield(s) the elemental form of **all** the constituent elements of the solute at the electrodes?

- A 3 only  
 B 1 and 2 only  
 C 1 and 3 only  
 D 2 and 3 only  
 E 1, 2 and 3

- 19** A redox reaction occurs in acidic solution between the complex ion  $\text{PtCl}_4^{2-}$  and the permanganate ion  $\text{MnO}_4^-$ . In this reaction:
- \* The platinum is oxidised from its state in the complex ion to the +4 oxidation state. \* The manganese is reduced from its state in the permanganate ion to the +2 oxidation state.
- What is the simplest whole number mole ratio of  $\text{PtCl}_4^{2-}$  to  $\text{MnO}_4^-$  required for the reaction?
- A** 1 : 1  
**B** 2 : 1  
**C** 2 : 5  
**D** 5 : 2  
**E** 5 : 1
- 20** A sample of a metal  $\text{M}$  consists of only two isotopes:  $^{69}\text{M}$  (relative isotopic mass 69.0) and  $^{71}\text{M}$  (relative isotopic mass 71.0).
- A 4.872 g sample of pure metal  $\text{M}$  reacts completely with oxygen to form 6.552 g of the oxide  $\text{M}_2\text{O}_3$ .
- What is the percentage abundance of the  $^{69}\text{M}$  isotope in this sample?
- (Assume the relative atomic mass of oxygen is 16.0.)
- A** 30%  
**B** 35%  
**C** 60%  
**D** 65%  
**E** 70%  
**F** 75%

- 21 Methanol can be synthesised industrially in the gas phase from carbon monoxide and hydrogen according to the following equation:



When 1 mol of  $\text{CO(g)}$  reacts completely with hydrogen, 112 kJ of energy is released.

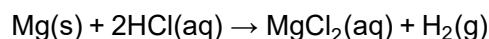
Some mean bond enthalpies are given in the table below:

| Bond         | Mean bond enthalpy / $\text{kJ mol}^{-1}$ |
|--------------|-------------------------------------------|
| $\text{C-H}$ | 413                                       |
| $\text{C-O}$ | 358                                       |
| $\text{O-H}$ | 464                                       |
| $\text{H-H}$ | 436                                       |

What is the bond enthalpy of the carbon–oxygen bond in the carbon monoxide molecule?

- A 613  $\text{kJ mol}^{-1}$
- B 965  $\text{kJ mol}^{-1}$
- C 1077  $\text{kJ mol}^{-1}$
- D 1189  $\text{kJ mol}^{-1}$
- E 1301  $\text{kJ mol}^{-1}$
- F 1513  $\text{kJ mol}^{-1}$

- 22 Magnesium reacts with hydrochloric acid according to the equation:



Five experiments are set up to produce hydrogen gas.

| Exp | Mass Mg / g | Vol HCl / $\text{cm}^3$ | Conc HCl / $\text{mol dm}^{-3}$ | State of Mg | Temp / $^{\circ}\text{C}$ |
|-----|-------------|-------------------------|---------------------------------|-------------|---------------------------|
| 1   | 1.2         | 200                     | 2.0                             | Powder      | 60                        |
| 2   | 3.6         | 150                     | 1.0                             | Powder      | 50                        |
| 3   | 2.4         | 300                     | 1.0                             | Ribbon      | 20                        |
| 4   | 2.4         | 100                     | 2.0                             | Powder      | 40                        |
| 5   | 4.8         | 100                     | 2.0                             | Ribbon      | 40                        |

(Ar value: Mg = 24)

Which experiment produces at least 0.10 mol of  $\text{H}_2$  in the shortest time?

- A Experiment 1
- B Experiment 2
- C Experiment 3
- D Experiment 4
- E Experiment 5

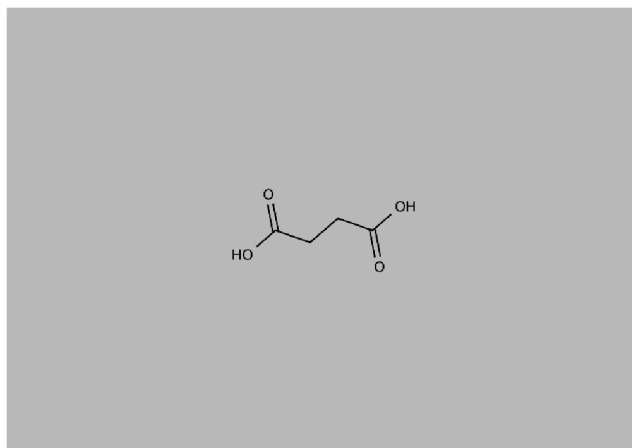
**23** An organic carboxylic acid **X** has the structural formula shown below:

A student dissolves 0.590 g of acid **X** in distilled water and makes the volume up to 250 cm<sup>3</sup> in a volumetric flask.

A 25.0 cm<sup>3</sup> portion of this solution is transferred to a conical flask and titrated against aqueous potassium hydroxide, KOH(aq). Exactly 20.0 cm<sup>3</sup> of the KOH(aq) solution is required for complete neutralisation.

What is the concentration of the potassium hydroxide solution in mol dm<sup>-3</sup>?

(A<sub>r</sub> values: H = 1.0, C = 12.0, O = 16.0)



- A** 0.0125 mol dm<sup>-3</sup>
- B** 0.0250 mol dm<sup>-3</sup>
- C** 0.0500 mol dm<sup>-3</sup>
- D** 0.100 mol dm<sup>-3</sup>
- E** 0.250 mol dm<sup>-3</sup>
- F** 0.500 mol dm<sup>-3</sup>

- 24 An addition polymer is formed from a hydrocarbon monomer  $\text{C}_6\text{H}_{12}$  with the molecular formula  $\text{C}_6\text{H}_{12}$ .

In the repeating unit of this polymer:

- the main chain (polymer backbone) consists of 2 carbon atoms
- every side group attached to the main chain is a methyl group ( $\text{CH}_3$ )
- neither of the two carbon atoms in the main chain is bonded to more than one hydrogen atom.

Which of the following is the systematic IUPAC name for  $\text{C}_6\text{H}_{12}$ ?

- A hex-1-ene  
 B hex-2-ene  
 C hex-3-ene  
 D 2-methylpent-2-ene  
 E 3-methylpent-2-ene  
 F 2,3-dimethylbut-1-ene  
 G 2,3-dimethylbut-2-ene  
 H 3,3-dimethylbut-1-ene

- 25 Element X is in Period 3 of the Periodic Table. In its ground state, an atom of X has three unpaired electrons, and it reacts with chlorine to form a chloride with the formula  $\text{XC}_l_3$ .

Which row of the table correctly describes the elements adjacent to X in the Periodic Table?

|   | element immediately above X                           | element immediately below X                    | element immediately to the left of X           | element immediately to the right of X       |
|---|-------------------------------------------------------|------------------------------------------------|------------------------------------------------|---------------------------------------------|
| A | exists as diatomic molecules containing a triple bond | has a larger atomic radius than X              | forms an oxide with a giant covalent structure | has a higher first ionisation energy than X |
| B | can expand its octet to form a pentachloride          | has a larger atomic radius than X              | contains one more proton in its nucleus than X | has a lower first ionisation energy than X  |
| C | exists as diatomic molecules containing a triple bond | has a larger atomic radius than X              | forms an oxide with a giant covalent structure | has a lower first ionisation energy than X  |
| D | has a lower first ionisation energy than X            | contains fewer occupied electron shells than X | forms an oxide with a giant covalent structure | has a higher first ionisation energy than X |
| E | exists as diatomic molecules containing a triple bond | has a higher first ionisation energy than X    | contains one more proton in its nucleus than X | has a lower first ionisation energy than X  |

- 26** An ion with the formula  $\text{Q}_4\text{O}_6^{2-}$  consists of a linear chain of four atoms of element Q. The two central Q atoms are bonded only to other Q atoms. The two terminal Q atoms are each bonded to one Q atom and three oxygen atoms.

Which option gives the oxidation states of the Q atoms in this ion?

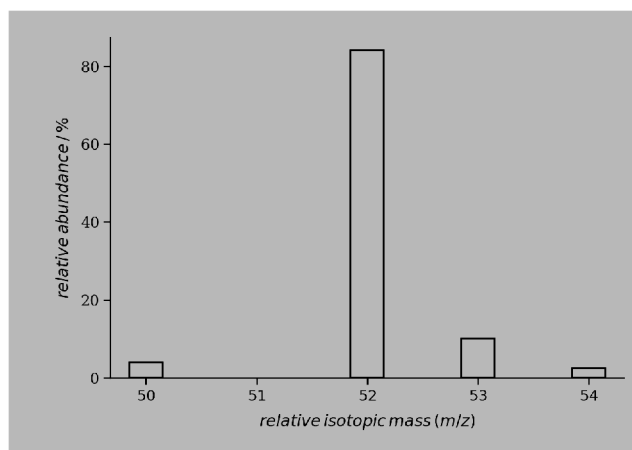
- A** +2.5 only
- B** 0 and +5
- C** 0 and +6
- D** +2 and +3
- E** +4 and +6

- 27** The mass spectrum of an element  $\text{X}$  is shown in the graph below.

A 0.100 mol sample of element  $\text{X}$  reacts completely with 0.075 mol of  $\text{O}_2(\text{g})$  to form a single oxide.

Naturally occurring oxygen consists of three isotopes with mass numbers 16, 17, and 18.

Using this information, what is the difference between the maximum possible and the minimum possible relative formula mass of this oxide?



- A** 6
- B** 8
- C** 10
- D** 12
- E** 14
- F** 16
- G** 22

## **Biology**

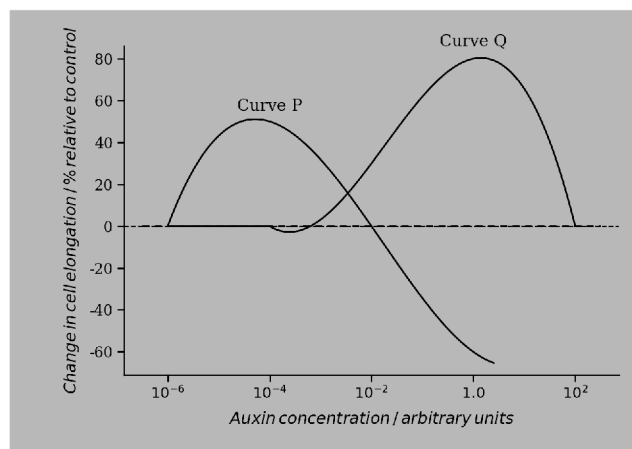
- 1 The graph shows the effect of auxin concentration on the elongation of cells in isolated root and shoot tissues of a plant species. One curve represents roots and the other represents shoots.

An intact seedling is placed horizontally in darkness. Due to gravity, auxin redistributes such that:

- the upper tissue layer in both the root and the shoot has an auxin concentration of  $10^{-4}$  arbitrary units
- the lower tissue layer in both the root and the shoot has an auxin concentration of 1.0 arbitrary units

Which row in the table correctly identifies the curve representing roots and predicts the resulting direction of curvature for the root and the shoot?

|   | Curve representing roots | Direction of root curvature | Direction of shoot curvature |
|---|--------------------------|-----------------------------|------------------------------|
| A | Curve P                  | downwards                   | upwards                      |
| B | Curve P                  | downwards                   | downwards                    |
| C | Curve P                  | upwards                     | upwards                      |
| D | Curve P                  | upwards                     | downwards                    |
| E | Curve Q                  | downwards                   | upwards                      |
| F | Curve Q                  | downwards                   | downwards                    |
| G | Curve Q                  | upwards                     | upwards                      |
| H | Curve Q                  | upwards                     | downwards                    |



- 2** Warm water at  $38^{\circ}\text{C}$  is continuously discharged from an industrial cooling system into a cold, slow-flowing river.

Which of the following could occur in the river downstream of the discharge point?

- 1 An increase in the mass of dissolved oxygen held in a given volume of water at saturation.
- 2 An increase in the rate of aerobic cellular respiration per unit body mass in resident ectothermic organisms.
- 3 A reduction in the population size of fish species with high dissolved oxygen requirements.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 3 The diagram shows a circular recombinant plasmid constructed by inserting a foreign gene of interest into a bacterial plasmid vector.

The original, non-recombinant plasmid contained three functional genes:

- gene A, coding for resistance to the antibiotic ampicillin
- gene B, coding for resistance to the antibiotic tetracycline
- gene C, coding for resistance to the antibiotic kanamycin

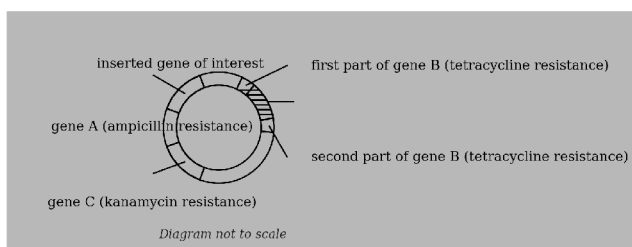
During the construction of the recombinant plasmid, the gene of interest was inserted into the coding region of gene B, as shown in the diagram.

Samples of bacteria containing only the recombinant plasmid and samples of bacteria containing only the original plasmid were separately inoculated onto four sets of agar plates:

- Plate 1: agar containing ampicillin and kanamycin
- Plate 2: agar containing ampicillin and tetracycline
- Plate 3: agar containing tetracycline and kanamycin
- Plate 4: nutrient agar containing no antibiotics

Which row in the table correctly identifies all the agar plates that will support the growth of each bacterial type?

|   | Agar plates supporting growth of bacteria containing only the recombinant plasmid | Agar plates supporting growth of bacteria containing only the original plasmid |
|---|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| A | 1 and 4 only                                                                      | 1, 2, 3, and 4                                                                 |
| B | 1 and 4 only                                                                      | 1, 2, and 3 only                                                               |
| C | 4 only                                                                            | 1, 2, 3, and 4                                                                 |
| D | 4 only                                                                            | 1, 2, and 3 only                                                               |
| E | 1, 2, and 4 only                                                                  | 1, 2, 3, and 4                                                                 |
| F | 2 and 3 only                                                                      | 1, 2, 3, and 4                                                                 |
| G | 1 only                                                                            | 1, 2, and 3 only                                                               |
| H | 1, 2, 3, and 4                                                                    | 1, 2, 3, and 4                                                                 |



- 4** Two strips of plant tissue, P and Q, are prepared for an experiment. The cells in Strip P have a mean internal solute concentration of  $0.6 \text{ mol dm}^{-3}$ . The cells in Strip Q have a mean internal solute concentration of  $0.2 \text{ mol dm}^{-3}$ .

Both strips are submerged in a beaker containing a sucrose solution with a concentration of  $0.4 \text{ mol dm}^{-3}$ .

Which option correctly predicts the initial change in turgor pressure for the cells in each strip?

- A** Strip P: Increases; Strip Q: Decreases
  - B** Strip P: Increases; Strip Q: Increases
  - C** Strip P: Decreases; Strip Q: Increases
  - D** Strip P: Decreases; Strip Q: Decreases
- 5** Which of the following statements about human cells is/are correct?
- 1 An osteocyte contains the same nuclear genes as a mesenchymal stem cell from the same individual.
  - 2 A mature spermatozoon contains half the number of chromosomes found in a secondary spermatocyte.
  - 3 When an adult stem cell divides asymmetrically, it produces one stem cell and one daughter cell that can undergo differentiation.
- (Assume that no mutations have occurred.)
- A** none of them
  - B** 1 only
  - C** 2 only
  - D** 3 only
  - E** 1 and 2 only
  - F** 1 and 3 only
  - G** 2 and 3 only
  - H** 1, 2 and 3

- 6 In a plant, tall height is controlled by a dominant allele  $T$  and dwarf height by a recessive allele  $t$ . A pure-breeding tall plant is crossed with a pure-breeding dwarf plant. Two offspring from this cross are then crossed.

What proportion of their offspring is expected to be dwarf?

- A 0  
B  $\frac{1}{8}$   
C  $\frac{1}{4}$   
D  $\frac{1}{2}$   
E  $\frac{3}{4}$   
F 1

- 7 The table shows the concentrations of substances in blood entering and leaving three different organs of a healthy adult who has been fasting for 24 hours:

- brain
- liver
- lungs

| organ | glucose /<br>mmol dm<br>(entering) <sup>g</sup> | oxygen /<br>arbitrary units<br>(entering) | urea /<br>mmol dm<br>(entering) <sup>g</sup> | glucose /<br>mmol dm<br>(leaving) <sup>-3</sup> | oxygen /<br>arbitrary<br>units<br>(leaving) | urea /<br>mmol dm<br>(leaving) <sup>-3</sup> |
|-------|-------------------------------------------------|-------------------------------------------|----------------------------------------------|-------------------------------------------------|---------------------------------------------|----------------------------------------------|
| 1     | 4.5                                             | 40                                        | 5.0                                          | 4.5                                             | 95                                          | 5.0                                          |
| 2     | 4.5                                             | 95                                        | 5.0                                          | 6.0                                             | 70                                          | 5.8                                          |
| 3     | 4.5                                             | 95                                        | 5.0                                          | 3.8                                             | 70                                          | 5.0                                          |

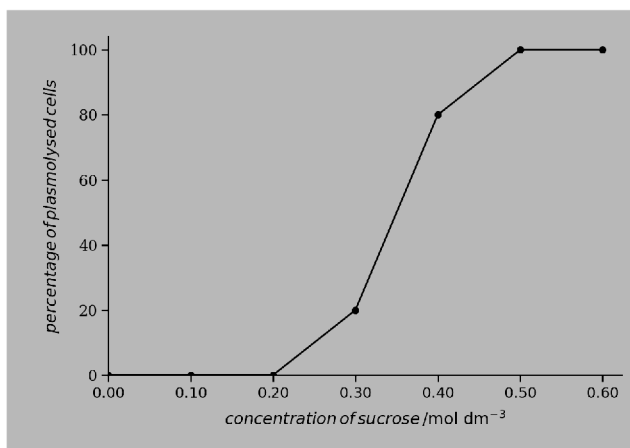
Which row in the table below correctly identifies the three organs?

|   | organ 1 | organ 2 | organ 3 |
|---|---------|---------|---------|
| A | lungs   | brain   | liver   |
| B | lungs   | liver   | brain   |
| C | liver   | lungs   | brain   |
| D | liver   | brain   | lungs   |
| E | brain   | lungs   | liver   |
| F | brain   | liver   | lungs   |

- 8 Strips of plant epidermal tissue were placed into sucrose solutions of different concentrations. After reaching equilibrium, the percentage of plasmolysed cells was determined for each solution. Plasmolysis occurs when a plant cell loses water and its cytoplasm pulls away from the cell wall.

The graph shows the results obtained.

Which option correctly describes the effect of increasing the concentration of sucrose from  $0.25 \text{ mol dm}^{-3}$  to  $0.40 \text{ mol dm}^{-3}$  on the percentage of plasmolysed cells, and states the underlying cause?

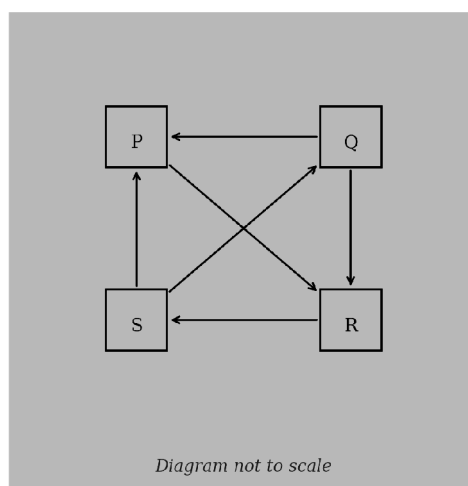


- A increases by 70% due to the net movement of water out of the cells
- B increases by 70% due to the net movement of sucrose into the cells
- C increases by 80% due to the net movement of water out of the cells
- D increases by 80% due to the net movement of sucrose into the cells
- E decreases by 70% due to the net movement of water into the cells
- F decreases by 70% due to the net movement of sucrose out of the cells
- G decreases by 80% due to the net movement of water into the cells
- H decreases by 80% due to the net movement of sucrose out of the cells

- 9 The diagram shows part of the terrestrial nitrogen cycle involving four pools of nitrogen labeled **P**, **Q**, **R**, and **S**.

Which row correctly identifies boxes **P**, **Q**, **R**, and **S**?

|          | Atmospheric<br>$\text{N}_2$ | Ammonium ions in<br>soil | Nitrate ions in<br>soil | Organic nitrogen compounds in<br>plants |
|----------|-----------------------------|--------------------------|-------------------------|-----------------------------------------|
| <b>A</b> | P                           | R                        | S                       | Q                                       |
| <b>B</b> | P                           | S                        | R                       | Q                                       |
| <b>C</b> | Q                           | S                        | R                       | P                                       |
| <b>D</b> | Q                           | R                        | S                       | P                                       |
| <b>E</b> | S                           | R                        | Q                       | P                                       |
| <b>F</b> | S                           | Q                        | R                       | P                                       |
| <b>G</b> | Q                           | P                        | S                       | R                                       |
| <b>H</b> | S                           | P                        | R                       | Q                                       |



- 10** Four distinct biological specimens (W, X, Y, and Z) were analysed. The results of the analysis are shown in the table below:

| Sample | Genome type                    | Enclosed by a phospholipid bilayer | Contains ribosomes | Capable of synthesizing ATP |
|--------|--------------------------------|------------------------------------|--------------------|-----------------------------|
| W      | double-stranded DNA (circular) | yes                                | yes (70S)          | yes                         |
| X      | single-stranded RNA            | yes                                | no                 | no                          |
| Y      | double-stranded DNA (linear)   | yes                                | yes (80S)          | yes                         |
| Z      | double-stranded DNA (linear)   | no                                 | no                 | no                          |

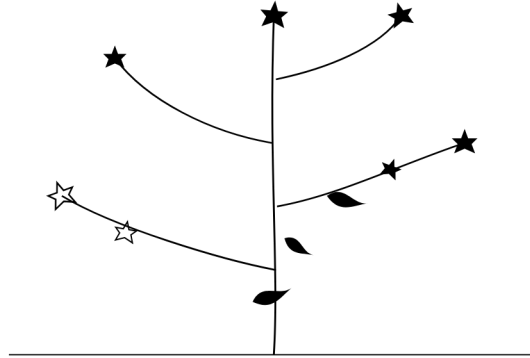
A student made the following statements about these specimens:

- 1 Sample X could be an enveloped virus, such as the influenza virus.
- 2 Sample W must be a mitochondrion and cannot be a free-living bacterium.
- 3 An antibiotic that selectively inhibits 70S ribosomes would directly inhibit protein synthesis in Sample W, but not in the cytoplasm of Sample Y.

Which of these statements is/are correct?

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 11** A seed is collected from a plant belonging to a true-breeding line that consistently produces red flowers. The seed is planted and grows into a new plant.
- This plant develops multiple branches, all of which bear red flowers, except for one single branch which exclusively bears white flowers, as shown in the diagram.



What is the most likely genetic explanation for the white-flowered branch?

- A** A somatic mutation occurred in a cell within the apical meristem of that specific branch.
- B** The parent plant was heterozygous, carrying a recessive allele for white flowers.
- C** The mutation for white flowers was present in the pollen grain that fertilised the parent flower.
- D** A localised deficiency of a specific mineral in the soil affected that branch.
- E** The mutation occurred in the zygote from which the entire plant grew.

- 12** In a species of mammal, coat pattern is determined by a single autosomal gene with two alleles,  $A^D$  and  $a$ .

- Embryos with the homozygous genotype  $A^D A^D$  do not survive and die before birth.
- Heterozygous individuals ( $A^D a$ ) have a striped coat.
- Homozygous individuals ( $aa$ ) have a solid-colour coat.

Two striped individuals are mated and produce a large number of live offspring.

Which of the following statements is/are correct?

- 1 The probability that any individual live-born offspring has a striped coat is  $\frac{2}{3}$ .
- 2 The ratio of allele  $A^D$  to allele  $a$  among the live-born offspring is 1 : 2.
- 3 If a striped live-born offspring is mated with a solid-colour individual, the probability of an embryo dying before birth is  $\frac{1}{4}$ .

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

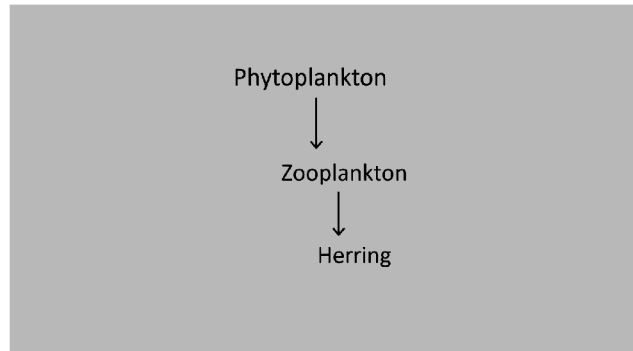
- 13** In honeybees (\*Apis mellifera\*), sex and fertility are determined by a single gene locus with multiple alleles ( $A_1, A_2, A_3$ , etc.) via complementary sex determination:

- **Haploid** individuals develop from unfertilised eggs, possess only one allele at this locus (hemizygous), and develop into **fertile males**.
- **Diploid** individuals develop from fertilised eggs:
  - if they are **heterozygous** at this locus, they develop into **fertile females**;
  - if they are **homozygous** at this locus, they develop into **sterile males**.

Which row in the table correctly shows the sex and fertility of the five honeybees with the given ploidy and genotypes?

|   | Individual 1<br>(Haploid, $A_1$ ) | Individual 2<br>(Diploid, $A_1 A_2$ ) | Individual 3<br>(Diploid, $A_1 A_1$ ) | Individual 4<br>(Diploid, $A_2 A_3$ ) | Individual 5<br>(Diploid, $A_3 A_3$ ) |
|---|-----------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|---------------------------------------|
| A | fertile male                      | fertile female                        | fertile female                        | fertile female                        | fertile female                        |
| B | fertile male                      | fertile female                        | fertile male                          | fertile female                        | fertile male                          |
| C | fertile male                      | fertile female                        | sterile female                        | fertile female                        | sterile female                        |
| D | fertile male                      | fertile female                        | sterile male                          | fertile female                        | fertile female                        |
| E | fertile male                      | fertile female                        | sterile male                          | fertile female                        | sterile male                          |
| F | fertile male                      | fertile female                        | sterile male                          | sterile female                        | sterile male                          |
| G | sterile male                      | fertile female                        | sterile male                          | fertile female                        | sterile male                          |
| H | sterile male                      | fertile female                        | fertile male                          | fertile female                        | fertile male                          |

- 14 The diagram shows a simplified aquatic food web.



Which of the following statements about this food web is/are correct?

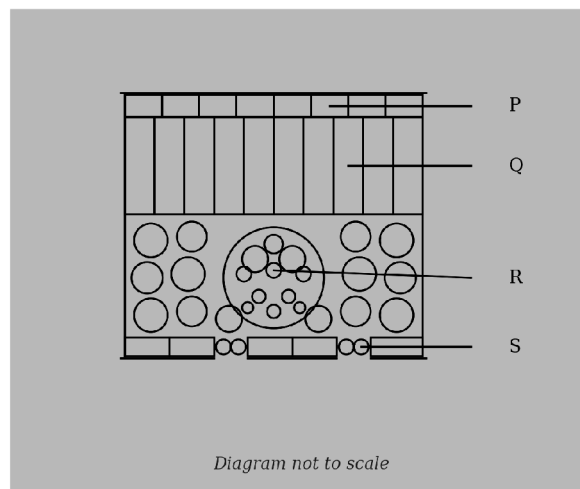
1. In this food web, the herring are primary consumers.
2. Approximately 90% of the energy from the zooplankton is transferred to the herring.
3. Decomposers make mineral ions from dead organic matter available to the phytoplankton.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 15** The diagram shows a transverse section through a dicotyledonous plant leaf with four distinct regions labelled P, Q, R, and S.

Which of the following statements is/are correct?

- 1 The entire leaf structure shown is classified as an organ.
- 2 Region P and region Q can each be classified as a tissue.
- 3 The mature xylem vessel elements in region R contain mitochondria to provide energy for the active transport of water.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 16** Natural selection leads to evolutionary change in a population when a selection pressure acts upon heritable phenotypic variation, resulting in differential reproductive success.

In which of the following scenarios will the environmental factor described act as a selection pressure that results in evolution by natural selection for the specified trait?

1 A population of clonal bacteria with identical genomes is exposed to an increased temperature within their survival range, where heat-shock protein production increases equally in all individuals via phenotypic plasticity.

2 A population of field mice with heritable variation in fur colour is introduced to a light sandy habitat where predatory owls hunt by sight.

3 A population of an annual plant in which variation in seed size is determined entirely by local soil depth with no genetic basis, growing in a region experiencing an increased frequency of summer droughts.

4 A population of pathogenic bacteria containing a rare allele encoding a membrane efflux pump, exposed to therapeutic concentrations of an antibiotic that is expelled by this pump.

- A** 1 and 2 only
- B** 2 and 4 only
- C** 3 and 4 only
- D** 1, 2 and 4 only
- E** 2, 3 and 4 only
- F** 1, 2, 3 and 4

- 17** Researchers investigated the function of Protein Q in mitochondria. They compared wild-type yeast cells with a mutant strain lacking Protein Q.

The table below shows the results of three measurements:

1. The rate of pyruvate oxidation (respiration) in intact mitochondria. 2. The activity of Krebs cycle enzymes measured in lysed mitochondrial extracts. 3. The percentage of soluble matrix proteins found in the cytoplasm (outside the mitochondria).

| Measurement                                                   | Wild-type | Mutant (lacking Protein Q) |
|---------------------------------------------------------------|-----------|----------------------------|
| Rate of pyruvate oxidation (arbitrary units)                  | 100       | 5                          |
| Activity of Krebs cycle enzymes in extracts (arbitrary units) | 100       | 98                         |
| Matrix proteins found in cytoplasm (%)                        | < 1       | 65                         |

Which conclusion regarding the role of Protein Q is best supported by these data?

- A** Protein Q is an enzyme that catalyses a rate-limiting step in the Krebs cycle.
- B** Protein Q is a transcription factor required for the expression of genes encoding respiratory enzymes.
- C** Protein Q is essential for maintaining the structural integrity of the mitochondrial membranes.
- D** Protein Q acts as a specific carrier protein for the active transport of pyruvate into the matrix.
- E** Protein Q is an inhibitor that prevents the breakdown of Krebs cycle enzymes in the cytoplasm.

- 18** A diploid eukaryotic cell is homozygous for a gene encoding an enzyme.

The cell undergoes mitosis. During DNA replication prior to cell division, a single base substitution mutation occurs on one of the newly synthesised chromatids at this gene locus.

The cell successfully completes mitosis to produce two daughter cells: Cell 1 (which inherits the mutated chromatid) and Cell 2 (which inherits only non-mutated chromatids).

Assume no other mutations or chromosomal alterations occur.

Which of the following statements is/are correct?

- 1 The total number of gene loci in the nucleus is the same in both daughter cells.
- 2 Cell 1 has a greater number of different alleles for this gene than Cell 2.
- 3 Following DNA replication in the next cell cycle, the four chromatids carrying this gene in Cell 1 will all have identical base sequences.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

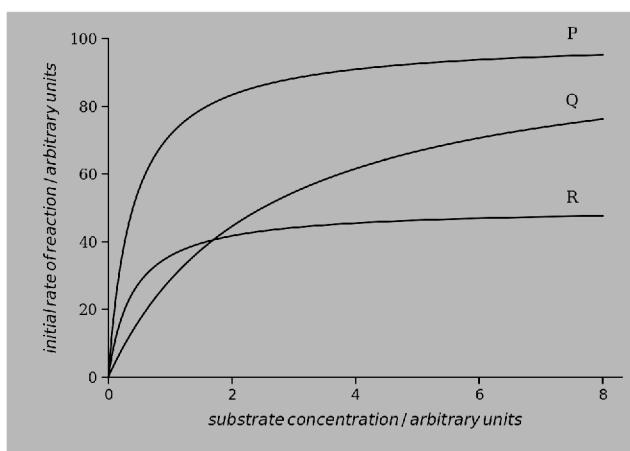
- 19** The graph shows the initial rate of an enzyme-catalysed reaction plotted against substrate concentration under three different experimental conditions: P, Q, and R.

In each experiment, the total concentration of enzyme present was identical.

- In experiment P, no inhibitor was added.
- In experiment Q, a constant concentration of substance X was added.
- In experiment R, a constant concentration of substance Y was added.

Which of the following statements is/are correct?

- 1 Substance X acts as a competitive inhibitor of the enzyme.
- 2 Increasing the substrate concentration can overcome the inhibitory effect of substance Y.
- 3 At a substrate concentration of 4 arbitrary units, a smaller proportion of enzyme molecules are bound to substrate in experiment Q than in experiment P.



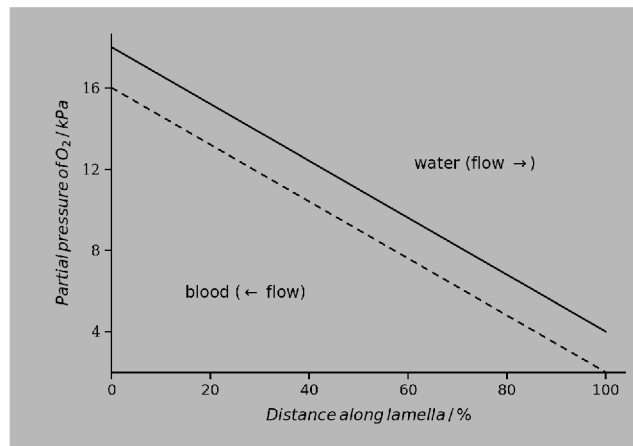
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 20** The graph shows the partial pressure of oxygen ( $p_{\text{ceO}_2}$ ) in water and in blood measured along the length of a secondary lamella in the gill of a bony fish.

Water flows across the lamella from distance 0 % to 100 %. Blood flows through the lamellar capillary in the opposite direction, from distance 100 % to 0 %.

Which of the following statements is/are correct?

- 1 Along the entire length of the lamella, net diffusion of oxygen occurs from the water into the blood.
- 2 Under the same flow conditions and entering partial pressures, if blood flowed in the same direction as water (concurrent flow), the maximum theoretical  $p_{\text{ceO}_2}$  the blood could reach is 10.0 kPa.
- 3 The partial pressure gradient driving the diffusion of oxygen across the lamella is greatest at distance 0 %.



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 21** A single-stranded RNA virus replicates inside host cells by first synthesising a complementary negative-sense RNA strand to form a double-stranded RNA intermediate.

The percentage of bases in the positive-sense single-stranded viral RNA genome was determined:

| Base         | Percentage in positive-sense RNA (%) |
|--------------|--------------------------------------|
| adenine (A)  | 32                                   |
| uracil (U)   | 18                                   |
| guanine (G)  | 28                                   |
| cytosine (C) | *w*                                  |

Which of the following statements is/are correct?

- 1 In the complementary negative-sense RNA strand, 18% of the bases are adenine.
- 2 In the double-stranded RNA intermediate, 22% of the bases are cytosine.
- 3 In the double-stranded RNA intermediate, 50% of the bases are purines.

- A** 1 only  
**B** 2 only  
**C** 3 only  
**D** 1 and 2 only  
**E** 1 and 3 only  
**F** 2 and 3 only  
**G** 1, 2 and 3  
**H** none of them

- 22** A double-stranded DNA molecule is 400 base pairs long and 32% of its bases are cytosine. One strand of this DNA molecule is transcribed to produce a single-stranded mRNA molecule that is 400 nucleotides long.
- In this mRNA molecule:
- 22% of the bases are guanine
  - 18% of the bases are uracil
- Which of the following statements is/are correct?
- 1 The mRNA molecule contains 168 cytosine bases.
  - 2 There are 1056 hydrogen bonds between the base pairs in the double-stranded DNA molecule.
  - 3 Adenine makes up 36% of the bases in the mRNA molecule.
- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 23** Three separate cultures of cells (Culture 1, Culture 2, and Culture 3) were established. At time  $t = 0$  h, each culture contained 100 stem cells and no specialised (differentiated) cells.

All stem cells in each culture divided synchronously every 24 hours according to a specific mode of division:

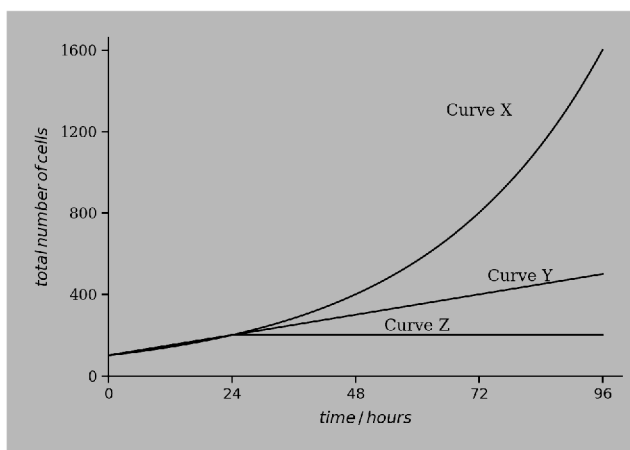
- **Culture 1:** each stem cell divides asymmetrically to produce one stem cell and one specialised cell.
- **Culture 2:** each stem cell divides symmetrically to produce two stem cells.
- **Culture 3:** each stem cell divides symmetrically to produce two specialised cells.

Specialised cells do not divide further, and no cells die during the 96-hour period.

The graph shows the total number of cells (stem cells plus specialised cells) over time for the three cultures, represented by curves X, Y, and Z.

Which row in the table correctly matches each culture to its corresponding curve on the graph, and gives the percentage of cells in Culture 1 that are stem cells at  $t = 96$  h?

|          | Culture 1 | Culture 2 | Culture 3 | Percentage of stem cells in Culture 1 at $t=96$ h |
|----------|-----------|-----------|-----------|---------------------------------------------------|
| <b>A</b> | Curve Y   | Curve X   | Curve Z   | 20%                                               |
| <b>B</b> | Curve Y   | Curve X   | Curve Z   | 25%                                               |
| <b>C</b> | Curve Y   | Curve Z   | Curve X   | 20%                                               |
| <b>D</b> | Curve Y   | Curve Z   | Curve X   | 50%                                               |
| <b>E</b> | Curve X   | Curve Y   | Curve Z   | 20%                                               |
| <b>F</b> | Curve X   | Curve Y   | Curve Z   | 25%                                               |
| <b>G</b> | Curve Z   | Curve X   | Curve Y   | 20%                                               |
| <b>H</b> | Curve Z   | Curve Y   | Curve X   | 50%                                               |

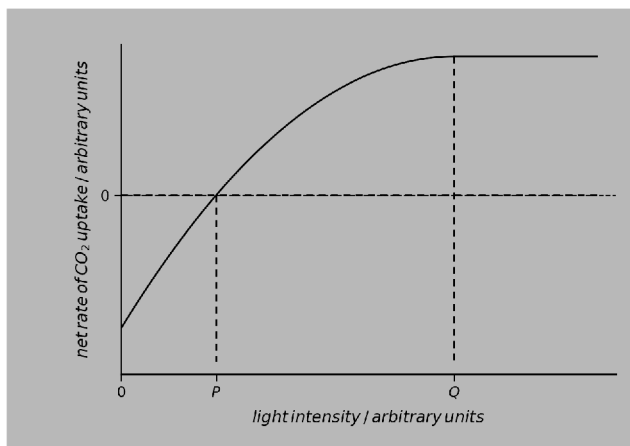


- 24** An experiment was carried out to investigate the effect of light intensity on the net rate of carbon dioxide ( $\text{CO}_2$ ) uptake by a plant leaf at a constant temperature of  $20^\circ\text{C}$  and constant atmospheric  $\text{CO}_2$  concentration. The graph shows the results obtained.

A student considered the graph and made the following conclusions:

- 1 Between light intensities 0 and  $P$ , no photosynthesis takes place.
- 2 At light intensity  $P$ , the rate of photosynthesis is equal to the rate of cellular respiration.
- 3 Between light intensities  $P$  and  $Q$ , light intensity is limiting the rate of photosynthesis.

Which of the student's conclusions is/are correct?



- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 25** A linear double-stranded DNA fragment contains 1500 base pairs and has a total of 3600 hydrogen bonds between its complementary nitrogenous bases.

Which of the following statements is/are correct?

- 1 Guanine accounts for 40% of the total nitrogenous bases in this DNA fragment.
- 2 There are 900 thymine bases in this DNA fragment.
- 3 There are 2998 phosphodiester bonds in the sugar–phosphate backbones of this DNA fragment.

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 26** Which of the following could be found inside a healthy pancreatic beta cell of an adult human?

- 1 circular DNA
- 2 gene for rhodopsin
- 3 cellulose

- A** none of them
- B** 1 only
- C** 2 only
- D** 3 only
- E** 1 and 2 only
- F** 1 and 3 only
- G** 2 and 3 only
- H** 1, 2 and 3

- 27** In a diploid organism, Process X produces cells required for growth and tissue repair, resulting in daughter cells that are genetically identical to the parent cell. Process Y produces cells required for sexual reproduction, resulting in daughter cells with half the number of chromosomes of the parent cell.

Which of the following statements correctly describes a difference in the chromosomal events between Process X and Process Y?

- A** DNA replication occurs prior to the start of Process X, but not prior to the start of Process Y.
- B** Sister chromatids separate and move to opposite poles in Process X, but not in Process Y.
- C** Homologous chromosomes pair up to form bivalents in Process Y, but not in Process X.
- D** Spindle fibres attach to the centromeres of chromosomes in Process Y, but not in Process X.
- E** Chromosomes condense and become visible in Process X, but not in Process Y.